Common Ownership and Worker Mobility*

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Abstract

We study how ownership links shape the mobility of human capital across independent organizations. We construct a novel dataset combining LinkedIn-based employment histories with venture capital (VC) investment data for U.S. ventures. Sharing a common investor connects firms that may otherwise compete for workers, but may also reduce matching frictions and facilitate worker reallocation across portfolio companies. We find that ventures exhibit higher job-to-job mobility after they start sharing an investor with another venture operating in the same local maket. These changes are larger when the shared investor holds a board seat, consistent with active governance as an important channel. Worker moves are disproportionately more likely to occur between firms connected through a shared investor, and separating workers are disproportionately absorbed by ventures in the shared investor's portfolio, with shorter transition times among workers redeployed locally. Overall, our results suggest that common ownership can create an internal labor market across formally independent ventures, facilitating hiring, worker reallocation, and reemployment after separation.

Keywords: common ownership; labor market; venture capital; competition; innovation

JEL Codes: J60, L20, L40, O31

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1. Introduction

Internal labor markets allow firms to reallocate workers across business units. Can similar arrangements operate across the boundaries of independent organizations? We examine this question by studying how common ownership shapes worker mobility.

Common ownership arises when investors hold stakes in multiple firms within the same industry and has become increasingly prevalent (Gutiérrez and Philippon 2016; Backus et al. 2021b), raising new questions about how overlapping financial interests shape firm behavior. A large literature shows that common ownership can soften product market rivalry and alter firms' innovation incentives (Azar et al. 2018; Antón et al. 2025), yet we know much less about its implications for the allocation of human capital, where the networks created by shared investors may influence how firms compete for, retain, and deploy workers. We address this gap by studying the venture capital sector. This sector is not only central to entrepreneurial growth and innovation (Kortum and Lerner 2000; Puri and Zarutskie 2012), but also one in which investors frequently hold stakes in multiple ventures within the same industry and geography (Eldar and Grennan 2021; Chen et al. 2010) and play active roles in monitoring and governance (Hellmann and Puri 2002; Bernstein et al. 2016; Gompers et al. 2020).

Understanding how common ownership affects the allocation of workers is important for assessing its broader implications for firm growth and competition, especially in settings where human capital is central to innovation (Stolpe 2002; Rosenkopf and Almeida 2003; Agrawal et al. 2006) and competitive advantage (Coff 1997; Lippman and Rumelt 1982). This question is also increasingly relevant for antitrust policy. Recent antitrust enforcement debates have placed greater emphasis on labor-market considerations (Shapiro 2019), while agencies have also expressed growing concern about common ownership and minority investments.[1] These concerns are reflected in the 2023 Merger Guidelines, which separately address mergers involving competing buyers in labor markets and acquisitions involving partial ownership or minority interests.[2]

The existing literature on common ownership points to a potential negative effect.

[1] For example, the Federal Trade Commission held a hearing on common ownership in 2018, and the European Commission examined the potential effects of common ownership in Dow/DuPont and Bayer/Monsanto (Burnside and Kidane 2020).

[2] Guideline #10 of the 2023 Merger Guidelines states: "When a Merger Involves Competing Buyers, the Agencies Examine Whether It May Substantially Lessen Competition for Workers, Creators, Suppliers, or Other Providers." Guideline #11 of the 2023 Merger Guidelines states: "When an Acquisition Involves Partial Ownership or Minority Interests, the Agencies Examine Its Impact on Competition." Guidelines: https://www.justice.gov/atr/merger-guidelines.

If managers maximize shareholders' value, the presence of investors with financial interests in multiple firms in the same industry may reduce firms' incentives to compete aggressively. In labor markets, this logic implies that common ownership may soften competition for workers: portfolio firms may internalize worker diversion across commonly owned employers, becoming less willing to poach employees from one another. Ownership links may also facilitate collusion through "no-poach" agreements.[3]

However, we highlight a countervailing channel through which common ownership can increase worker mobility: common investors may act as labor-market intermediaries. Even without direct control over hiring decisions, shared investors can transmit information about job opportunities and refer talent across portfolio companies, thereby reducing search and matching frictions across otherwise independent firms. In this sense, common VC ownership may create an internal labor market across formally independent ventures, analogous to the way large organizations reallocate workers across units (Baker and Holmstrom 1995). This mechanism is visible in the recruiting infrastructure of the venture capital industry, where many VC firms maintain portfolio-wide job boards and talent networks that aggregate vacancies across independent portfolio companies.[4] Therefore, the net effect of common ownership on worker mobility is theoretically ambiguous.

To provide evidence on how common ownership affects worker mobility, we combine comprehensive venture capital (VC) investment data from PitchBook and VentureXpert with micro-level data on employee movements from Revelio Labs to construct a panel of ventures operating across industries and Core Based Statistical Areas (CBSAs). We focus on high-growth ventures that raised VC financing between 1990 and 2021, and collect information on their state of incorporation from VentureXpert and OpenCorporates.

We first examine whether ventures that start sharing investors with other ventures active in the same industry–CBSA (henceforth, "a local market") exhibit changes in worker mobility. Becoming commonly owned is associated with an approximate 5.8 percentage-point increase in the likelihood of a job-to-job (J2J) mobility event, defined as an employee leaving the firm and starting a new job within one year in the same local market. This association is economically large. These patterns are common across different roles: J2J flows increase across all seniority levels, being especially pronounced among scientists and engineers.

[3]For instance, in the early 2000s, several Silicon Valley firms entered into agreements not to recruit each other's employees (Gibson 2024). Herrera-Caicedo et al. (2025) shows that such agreements were associated with firms sharing a common leader, such as a board member.

[4]As an example, Figure C.1 reports a screenshot of Andreessen Horowitz's portfolio jobs board. Although the public components of these platforms are accessible to workers outside the portfolio, they illustrate that investors actively organize labor-market information around their portfolio firms.

To address potential endogeneity concerns arising from unobservable local-market, time-varying shocks that may affect both investors' incentives to engage in common ownership and firm labor outcomes, we exploit variation from the staggered adoption of corporate opportunity waiver (COW) statutes across nine U.S. states between 2000 and 2016. These laws authorize firms incorporated in adopting states to include charter provisions that release directors and investors from liability when they pursue business opportunities that could otherwise be viewed as belonging to the firm (e.g., Rauterberg and Talley 2017). Because VCs often hold board seats and have access to sensitive information about their portfolio companies, investing in multiple related ventures can expose them to potential conflicts of interest. By limiting directors' and investors' exposure to this litigation risk, COW statutes reduce legal frictions associated with investing in multiple firms within the same local market. Importantly, Eldar and Grennan (2024) provides evidence that COW adoption was not driven by lobbying efforts or broader economic trends, was narrowly targeted at limiting fiduciary-duty liability, and was unrelated to incorporation choices, reducing the concern that these statutes affect venture outcomes through channels other than common ownership.

We implement a stacked difference-in-differences design (Cengiz et al. 2019), constructing cohort-specific datasets around each COW law passage and estimating treatment effects within each cohort window. This design addresses the heterogeneity-induced biases that can affect conventional two-way fixed-effects DiD when treatment timing is staggered (Goodman-Bacon 2021). In the ten years following COW adoption, firms incorporated in adopting states experience a 1.42 percentage-point increase in the probability of worker mobility. Because COW statutes expand investors' discretion to favor one portfolio company over another when conflicts of interest arise, the increase in common ownership suggests that venture capital investors deliberately create ownership structures that enable active management across portfolio firms. We also follow Eldar and Grennan (2024) and instrument for common VC ownership with a venture's incorporation in a COW-adopting state, with qualitatively similar results.

Two sets of analyses help interpret the economic forces behind our findings. First, we analyze whether the effects of common ownership vary with the investors' ability to influence the governance of portfolio firms. If common investors facilitate labor reallocation by transmitting information, coordinating hiring, or actively connecting portfolio firms, their effects should be stronger when they have board representation. We find larger changes in mobility when the shared investors sit on the board, consistent with an active governance channel. Second, we examine whether the documented

patterns vary with concentration in the VC-backed segment of the local labor market. If common ownership softens competition for workers, such effects should be more pronounced where workers have fewer outside options and commonly owned firms tend to account for a large share of local employment. Measuring concentration with the Herfindahl–Hirschman Index (HHI) based on firms' employment shares within each industry–CBSA up to the year in which common ownership arises, we show that changes in worker mobility are systematically attenuated, but remain positive, in more concentrated local markets.

We then directly examine the extent to which ownership links across firms generated by VC investments contribute to the creation of an internal labor market through which investor connections help organize where workers move. We trace worker flows between ventures located in the same CBSA and examine whether they follow ownership links. Ventures exchange workers more frequently when they share a VC investor, particularly within industries, where skills are likely to be more transferable. This pattern extends to scientists and engineers, as well as C-suite executives, whose specialized knowledge may make information about worker capabilities especially valuable. Worker exchange also increases following the first investment from a shared investor, both between similarly staged ventures and from earlier-stage origins to later-stage destinations. These patterns suggest that investor networks connect firms with potentially different personnel needs across the venture development process.

A central function of internal labor markets is to accommodate shocks to business units's labor demand by facilitating worker redeployment. To examine this role, we first follow individual separations and study which employers absorb departing workers and how quickly these transitions occur. Separating workers disproportionately join ventures connected to their previous employer through a shared investor, and, among workers reemployed locally, transitions to connected firms occur more quickly. Then, we study whether investor networks provide a channel to redeploy workers following contractions or liquidity events affecting a portfolio firm. We find that workers leaving sharply contracting ventures disproportionately join ventures connected through a shared investor. Similarly, after an IPO, workers leaving the newly public company disproportionately join connected portfolio private ventures.

Finally, because the benefits of worker redeployment within investor portfolios depend on the quality of the resulting matches, we follow workers at their new employers and find that connected moves are associated with broadly similar retention and faster promotions. These patterns suggest that investor networks facilitate timely worker real-

location without an apparent reduction in match quality.

Overall, our results show that common ownership has effects that extend beyond product-market competition: in entrepreneurial labor markets, shared investors can function as intermediaries that facilitate worker mobility across independent organizations.

Related Literature. Our analyses contribute to a large literature examining the effects of common ownership of public firms by institutional investors on prices (Azar et al. 2018), entry (Newham et al. 2025), innovation (Antón et al. 2025), and overall welfare (Backus et al. 2021a; Ederer and Pellegrino 2026). Closer to our work, Azar et al. (2022) exploits a firm's inclusion in the S&P 500 index as a shock to the common ownership of its competitors in local labor markets. They conclude that common ownership reduces average annual earnings per employee and raises separation rates, but ultimately increases total employment in the local labor market.[5]

We complement this line of research by studying the VC industry, a different institutional setting characterized by the presence of investors with more significant control rights (Gompers et al. 2020) and by the existence of clear formal and informal mechanisms through which investors can monitor and influence the managerial strategy of their portfolio firms—such as the appointment of board representatives (Amornsiripanitch et al. 2019; Ewens and Malenko 2020).[6] In this context, we provide evidence of novel outcomes shaped by common ownership. By doing so, we also contribute to an emerging literature on common ownership in the VC industry, which has documented its effects on firms' ability to raise VC, exit rates, innovation efficiency, and the reallocation of financial and innovation resources within the portfolio (González-Uribe 2020; Li et al. 2023; Eldar and Grennan 2024; Leccese 2024).

Moreover, our paper relates to the literature on internal labor markets and labor reallocation within ownership networks. Prior work shows that conglomerates and business groups use internal labor markets to reallocate workers across organizational units or affiliated firms in response to shocks and changing business conditions, with implications for productivity and competitive advantage (Giroud and Mueller 2015; Tate and Yang 2015; Belenzon and Tsolmon 2016; Faccio and O'Brien 2021; Huneeus et al. 2021; Adhvaryu et al. 2024; Cestone et al. 2024; Tate and Yang 2024). Unlike conglomerates or business groups, commonly owned ventures do not typically share a common balance

[5] Relatedly, Emmens et al. (2024) studies how common ownership affects firms' automation decisions and the implications for employment. Moreover, Antón et al. (2023) shows that common ownership can shape managerial contracts, making them less performance-sensitive.

[6] A distinct but related literature studies the product market implications of interlocking directorates, highlighting how governance links across firms can affect competitive conduct (e.g., Barone et al. 2025).

sheet, integrated human-resources departments, or a controlling shareholder. Instead, they are independent firms connected through investors that hold minority stakes but may possess information about portfolio firms' labor needs and workers' skills. Our results show that such weaker ownership links can nevertheless generate labor-market intermediation, increasing within-portfolio flows and facilitating worker redeployment across portfolio firms following contractions or exits. In doing so, we show that internal-labor-market forces can arise not only inside conglomerates or business groups, but also in investor networks connecting independent firms.

Labor-market outcomes have also become central in antitrust. A firts strand of research studies on collusive practices, such as no-poach and wage-fixing agreements, which restrict worker mobility and suppress wages (Krueger and Ashenfelter 2022; Callaci et al. 2024; Lafontaine et al. 2025; Delabastita and Rubens 2025). Particularly relevant to our setting, where VCs often hold board seats, Herrera-Caicedo et al. (2025) show that sharing a common leader, such as a board member, increases the likelihood of entering a no-poach agreement. A second strand focuses on the effects of mergers on labor markets, showing that consolidation can depress wages, reduce match quality, and sometimes lead to layoffs (Arnold 2019; Prager and Schmitt 2021; Montag 2023; Shi 2024). We contribute to this literature by focusing on a different form of consolidation—common ownership—which has gained traction in antitrust debates, and examine its implications for worker mobility, a key driver of knowledge diffusion and innovation (Dasaratha 2023).[7]

2. Data

We assemble a novel dataset that combines venture capital investment, workers' employment trajectories, and legal incorporation records. This section describes the data sources, the procedure used to merge them, the construction of the main variables, and some summary statistics.

[7] A distinct body of work studies how non-compete agreements shape labor market outcomes, consistently finding that stricter enforceability reduces mobility and suppresses wage growth (Marx et al. 2009; Balasubramanian et al. 2022; Johnson et al. 2025). Common ownership is an alternative channel influencing similar outcomes in the VC ecosystem, where investor coordination rather than contractual restrictions governs worker reallocation.

2.1. Data sources

Our primary data on venture capital investments comes from PitchBook and VentureXpert (LSEG Data & Analytics). Both datasets report detailed information on financing rounds, including the identity of the investee and investors, the date and type of each round, the investee's headquarters ZIP code, and an industry classification. PitchBook assigns each venture to one of 41 industries, while VentureXpert uses a related but non-identical proprietary taxonomy. Because our measure of common ownership relies on consistent industry definitions, we manually construct a crosswalk between the two taxonomies (Appendix Table B.1).[8] To assign ventures to local labor markets, we map headquarters ZIP codes to Core Based Statistical Areas (CBSAs) using ZIP-code–level definitions from the U.S. Census Bureau.[9]

Data on employment, job transitions, and worker characteristics come from Revelio Labs, which extracts individual employment histories from LinkedIn profiles. Revelio provides start and end dates for each position, worker moves across employers, and educational and skill information. These data have been widely used in recent work, including studies of labor markets, firm dynamics, and entrepreneurship.[10]

Our identification strategy uses the staggered adoption of corporate opportunity waivers across U.S. states, which requires reliable information on ventures' incorporation states. We obtain this information from VentureXpert and from OpenCorporates, the largest global open database of legal entities. OpenCorporates provides firm-level registration and incorporation records.[11]

2.2. Sample construction and key variables

Our dataset combines venture investment histories, worker-level employment data, and incorporation records. We begin with the universe of ventures that raised at least one VC round between 1990 and 2021 in either PitchBook or VentureXpert. Because PitchBook provides rich recent coverage while VentureXpert offers substantially deeper pre-2007 coverage, we merge the two sources to obtain the most comprehensive longitudinal

[8] We rely on PitchBook's "Industry Group" and VentureXpert's "Primary Industry Sub-Group" in the baseline analyses and show robustness to using more granular classifications.

[9] This data is available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html.

[10] See, for example, Hacamo and Kleiner (2022), Bernstein et al. (2023), Jeffers (2023), Amanzadeh et al. (2024), Chen et al. (2025), Kwan et al. (2025).

[11] We gratefully acknowledge OpenCorporates for providing free academic access through its Public Benefit Project; data are available at https://opencorporates.com/use-cases/academia.

dataset available for VC-backed ventures.

We treat PitchBook as the primary dataset and merge in venture and investor records from VentureXpert. Ventures appearing in both sources are linked using fuzzy matching on firm names combined with exact matching on headquarters state. Ventures found only in VentureXpert are retained to expand historical coverage. Investor identities are matched by fuzzy name matching and exact matching on investor location to ensure consistency in venture–investor relationships.

To link ventures to worker-level employment histories, we follow a two-stage procedure. First, when PitchBook reports a LinkedIn URL, we match firms to Revelio using the exact URL. Second, for firms without a LinkedIn URL, we employ fuzzy matching on firm names combined with exact matching on headquarters state.

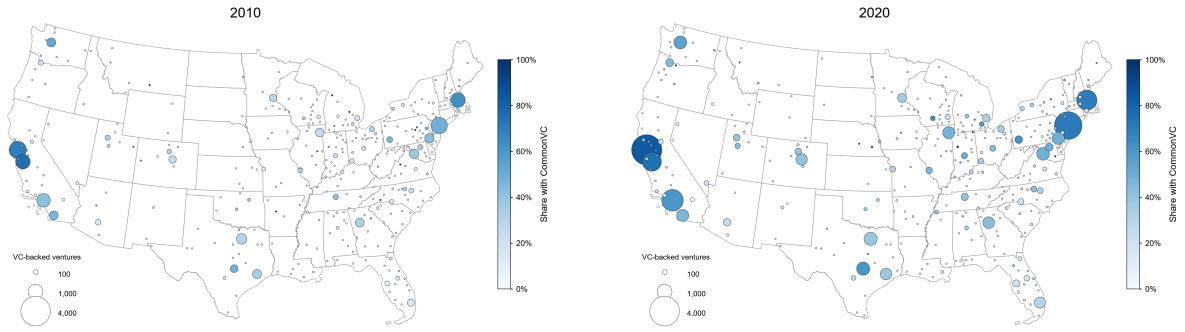
The merged dataset integrates (i) venture characteristics and investment histories from PitchBook and VentureXpert, (ii) worker mobility and employment outcomes from Revelio Labs, and (iii) incorporation-state information from VentureXpert and OpenCorporates.

Final Sample. We construct our firm–year panel considering all years in which a venture is active as a private firm. A venture exits the sample when it undergoes an IPO or when acquired. When these dates are unavailable, we follow Li et al. (2023) and assume that a venture is written off from an investor's balance sheet if it receives no new financing for five years after its last round. The year in which a venture enters our sample is the founding year when reported, and the year of first investment otherwise. The final sample includes 70,551 ventures across 39 industries and 567 CBSA.

Our measure of common ownership is a binary indicator, *Common VC*, equal to one if venture i shares at least one VC investor with another active venture in the same CBSA–industry–year.

Common ownership has expanded broadly across U.S. local labor markets. Figure 1 maps the share of VC-backed ventures with common ownership across U.S. CBSAs in 2010 (Panel A) and 2020 (Panel B). In 2010, common ownership was already present in a wide range of CBSAs—including established entrepreneurial hubs such as the San Francisco Bay Area, Boston, and New York, but also smaller markets in the Midwest, South, and Mountain West. By 2020, common ownership has expanded substantially, with meaningful shares now visible across most of the country. While the highest concentrations of common ownership remain in major hubs (with within-local market shares reaching 30–40%), the phenomenon is geographically widespread: approximately half of the ventures in our sample experience common ownership at some point during

FIGURE 1. Common ownership U.S. CBSAs, 2010 vs. 2020



A. 2010 B. 2020

Notes: Each circle represents a CBSA; size indicates the number of VC-backed ventures and shading reflects the share of those ventures with at least one common-VC investor.

their life-cycle within our sample.

Using Revelio data, we capture worker mobility in the firm-year panel, by counting workers who leave the focal firm and join another employer within the same local market in the same or following year. We construct $\mathbb{1}\{J2J\}$ as a binary variable equal to one whenever a firm-year exhibits a positive count. For example, if a worker employed at a biotech venture in Boston leaves the firm and joins another biotech venture in Boston within a year, we classify the move as a J2J. By contrast, if the same worker moves to a biotech firm in New Jersey or to a non-biotech firm in Boston, the move is not coded as a J2J in the firm-year panel.

Table 1 reports summary statistics. The average venture in our sample employs 53 workers and experiences a 10% turnover rate per year. Moreover, a venture has, on average, 1.2 investors, a cumulative investment of approximately $13.7 million, and is 19 years old at the time of observation. A nontrivial fraction of firms exhibit common ownership: 19.4% of firm-years feature at least one common investor; this figure falls to roughly 3% of firm-years when conditioning on common investors who also sit on the firm's board.[12]

Worker mobility is sizable on the extensive margin—48% of firm-years record at least one job-to-job transition—but transitions within the same industry-CBSA occur in only 5% of firm-years. We additionally examine the mobility of two worker categories of

[12]Figure C.2 compares the distribution of cumulative VC investment for firm-years where common ownership is observed against all the others, restricting attention to observations with positive cumulative VC investment. The distributions are similar, which is consistent with the fact that common ownership is not simply concentrated among the most heavily funded ventures.

TABLE 1. Summary statistics

	Obs	Mean	SD	p10	p50	p90
Common VC	862,899	0.194	0.396	0.00	0.00	1.00
Common VC with Board Seat	862,899	0.0287	0.167	0.00	0.00	0.00
Employment	862,899	52.743	142.819	1.00	10.00	111.00
Turnover Rate	862,899	0.0992	0.175	0.00	0.00	0.27
$\mathbb{1}$\{Overall J2J\}	862,899	0.482	0.500	0.00	0.00	1.00
$\mathbb{1}$\{J2J\}	862,899	0.0514	0.221	0.00	0.00	0.00
J2J count	862,899	0.125	1.117	0.00	0.00	0.00
$\mathbb{1}$\{Innovator J2J\}	862,899	0.128	0.334	0.00	0.00	1.00
$\mathbb{1}$\{Innovator J2J\}	862,899	0.0108	0.103	0.00	0.00	0.00
Innovator J2J count	862,899	0.0170	0.244	0.00	0.00	0.00
$\mathbb{1}$\{C-Suite J2J\}	862,899	0.00176	0.0420	0.00	0.00	0.00
C-Suite J2J count	862,899	0.00196	0.0487	0.00	0.00	0.00
$\mathbb{1}$\{Junior J2J\}	862,899	0.0300	0.171	0.00	0.00	0.00
Junior J2J count	862,899	0.0546	0.540	0.00	0.00	0.00
$\mathbb{1}$\{Mid-Seniority J2J\}	862,899	0.0165	0.127	0.00	0.00	0.00
Mid-Seniority J2J count	862,899	0.0265	0.305	0.00	0.00	0.00
$\mathbb{1}$\{Senior J2J\}	862,899	0.00269	0.0518	0.00	0.00	0.00
Senior J2J count	862,899	0.00303	0.0632	0.00	0.00	0.00
C-Suite Move	862,899	0.00272	0.0521	0.00	0.00	0.00
Cumulative VC Investment	862,899	13.657	55.264	0.00	0.00	23.90
VC Investor Count	862,899	1.242	2.237	0.00	0.00	4.00
Firm Age	862,899	18.830	21.484	2.00	11.00	46.00

Notes: This table presents summary statistics at the firm-year level. Employment is the number of workers with a position at the firm in the year. Turnover is spell-based: workers whose Revelio spell at the firm ends in the year and who have no next spell (before 2021) or a next spell at another employer, divided by the average of employment in the year and in the firm's next observed year (zero when there is none). $\mathbb{1}$\{Overall J2J\} when a worker's main job at the firm ends and the next main job, at another employer, starts within 12 months; a J2J move occurs when the destination is a venture in the same industry and CBSA. A move counts for a worker type when the worker's most senior position is of that type both at the origin in the exit month and at the destination in the month the job becomes the main job: innovators are scientists and engineers, and junior, mid-seniority and senior workers hold Revelio seniority levels 1–3, 4–5 and 6–7. The counts are the number of workers with such a move in the year. Common VC with Board Seat is one if a shared investor holds a board seat. C-Suite Move equals one if an executive recorded in PitchBook with a C-suite role moves between the firm and another sample venture (in either direction), dated by the last year at the origin, whatever the gap. Cumulative VC Investment is the firm's cumulative disclosed VC and PE round amounts in millions of dollars (PitchBook deal sizes; VentureXpert disclosed round amounts where PitchBook has none); VC Investor Count is the cumulative number of distinct VC and PE investors of the firm (PitchBook; VentureXpert where PitchBook has none); Firm Age is the year minus the founding year plus one. Employment, Cumulative VC Investment, VC Investor Count and Firm Age are winsorized at the 1st and 99th percentiles.

particular interest: scientists and engineers (which we label as "innovators"), and senior executives (C-Suites). These categories are substantively important because innovators carry technical knowledge central to entrepreneurial ventures, while executives shape firm strategy and governance. They also help address a potential data-coverage concern: while LinkedIn-based data may underrepresent low-skill workers who maintain less active profiles, coverage is particularly strong among skilled employees—the workers who constitute the bulk of the labor force in high-growth, innovation-driven ventures and whose mobility is most relevant for our analysis. In our sample, 1.1% of firm-years record an innovator transition and 0.2% record a C-suite move.

3. Common VC Ownership and the Mobility of Workers

We estimate the effect of common VC ownership on job-to-job mobility within a local market using three complementary empirical approaches. We first estimate a firm-level panel specification that exploits within-firm changes in common VC exposure. We then use the staggered adoption of Corporate Opportunity Waiver (COW) statutes as a source of plausibly exogenous variation in the legal costs associated with investing in related firms. After having analyzed the implications of COW statutes in a stacked difference-in-differences (DiD) design, we use COW eligibility as an instrument for common VC ownership.

3.1. Firm-level panel specification

For each firm j operating in industry n and CBSA c in year t, we use ordinary least squares (OLS) to estimate:

$$1\{J2J\}_{j,n,c,t} = \mu_j + \alpha_{c,t} + \gamma_{n,t} + \beta \cdot \text{CommonVC}_{j,t} + \mathbf{X}'_{j,t}\Lambda + \varepsilon_{j,n,c,t}, \tag{1}$$

where the outcome is an indicator for any within-market J2J move. $\text{CommonVC}_{j,t}$ equals one if firm j shares at least one VC investor with another firm in the same industry-CBSA in year t. Firm fixed effects, μ_j, absorb time-invariant differences across firms. CBSA-year fixed effects, $\alpha_{c,t}$, absorb time-varying local shocks, while industry-year fixed effects, $\gamma_{n,t}$, absorb shocks common to firms in the same industry and year. The vector $\mathbf{X}_{j,t}$ includes controls for investor characteristics Standard errors are clustered at the industry-CBSA level.

Table 2 presents our baseline estimates. Column 1 shows that sharing a VC investor

TABLE 2. Common ownership and within-market mobility

	(1) All workers	(2) Innovators	(3) C-suite	(4) Junior	(5) Mid-seniority	(6) Senior
CommonVC	0.0580***	0.0117***	0.00222***	0.0307***	0.0192***	0.00333***
	(0.00338)	(0.00124)	(0.000308)	(0.00188)	(0.00125)	(0.000459)
Observations	858,486	858,486	858,486	858,486	858,486	858,486
Adj. R-squared	0.271	0.158	0.015	0.199	0.163	0.032
Mean DV	0.0517	0.0109	0.00177	0.0302	0.0166	0.00270
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
CBSA-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: OLS estimates of firm-year regressions of the internal J2J indicator on *CommonVC*. The dependent variable equals one if at least one worker of the column's type makes an internal J2J move in the year (definitions in Table 1). *CommonVC* equals one if the firm shares a VC investor with another venture in the same industry and CBSA. Investor controls are the average age, assets under management, number of funds and number of IPOs of the firm's prior VC investors. Standard errors clustered at the industry-CBSA level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

with another venture in the same local market is associated with a 5.8-percentage-point higher probability that at least one employee leaves and starts a new job at another venture in that market within a year. As an illustrative benchmark, the complementary count estimates correspond to approximately one additional within-market move per year across eight to ten average-sized ventures.[13] These estimates are consistent with common ownership being associated with greater labor-market dynamism, rather than worker retention or labor-market foreclosure.

An important question, however, is whether these changes are concentrated among particular categories of workers. This distinction matters because the movement of technical and managerial talent is likely to have different implications for firm performance, innovation, and organizational change. The remaining columns in Table 2 therefore examines heterogeneity across worker types. Mobility rises throughout the organizational hierarchy, with statistically significant increases for innovators, junior employees, middle-management personnel, and senior employees, including C-suites. Although the largest increase in mobility is observed among junior workers in absolute terms, the

[13]The PPML coefficient in Table C.2 is 0.719, corresponding to approximately twice the expected number of moves ($e^{0.719} \approx 2.05$). Applying this multiplier illustratively to the full-sample mean of 0.125 moves per firm-year yields about 0.13 additional moves per firm-year. This is a scale comparison, not an estimated average treatment effect.

baseline mobility rate is considerably lower for senior personnel, implying even larger relative changes at the top of the organization.

3.2. Corporate opportunity waiver laws and common VC ownership

Equation (1) exploits within-firm variation in common ownership while flexibly controlling for local and industry-specific shocks. However, it remains subject to endogeneity concerns. Venture capitalists financing multiple firms in the same industry–CBSA may select ventures with different growth prospects, different risk profiles, or recent unobservable shocks that also affect hiring and worker mobility. We therefore complement this specification with variation from the staggered adoption of Corporate Opportunity Waiver (COW) statutes. We describe our approach below.

COW statutes allow firms to waive the corporate opportunity doctrine, reducing the risk that fiduciaries face liability when they pursue business opportunities that may overlap with those of another firm. In the venture capital setting, these statutes reduce legal uncertainty for investors and directors involved with multiple firms operating in related business areas. These laws were implemented by nine U.S. states in a staggered fashion between 2000 and 2016. Table C.1 provides details on these amendments.

Our empirical strategy builds on Eldar and Grennan (2024), which provides extensive qualitative and quantitative institutional evidence on the origins and implications of COW statutes. They show that the primary motivation for the adoption of COWs was to eliminate legal uncertainty surrounding fiduciary duties, rather than to respond to lobbying or broader economic trends. Their analyses further indicate that these laws had the main objective to shield investors and directors from liability when managing or investing in multiple firms with overlapping business opportunities. These institutional features mitigate concerns that COW statutes directly affected employment or worker mobility through channels unrelated to common ownership. Moreover, they survey venture lawyers to analyze the extent to which state incorporation decisions are endogenous. Specifically, while one may be concerned that high-growth ventures are more likely to incorporate in Delaware, Eldar and Grennan (2024) shows that incorporation decisions are largely driven by the familiarity of legal counsel with Delaware law and not by differences in substantive provisions such as COW statutes.

We use a stacked DiD design (Cengiz et al. 2019) to estimate the effect of COW adoption on the prevalence of common VC ownership and worker mobility. This approach addresses concerns that arise in conventional two-way fixed-effects estimators with staggered treatment timing (Goodman-Bacon 2021; De Chaisemartin and d'Haultfoeuille

2020). For each adoption cohort $g \in \{2000, 2001, 2003, 2005, 2006, 2007, 2011, 2014, 2016\}$, corresponding to the year in which a state adopted a COW statute, we construct a cohort-specific dataset containing treated firms incorporated in the adopting state and control firms incorporated in states that never adopted COW statutes. We observe firms in a symmetric window from five years before to ten years after COW adoption, $[g-5, g+10]$, and then stack the cohort-specific datasets.

We estimate the following event-study specification:

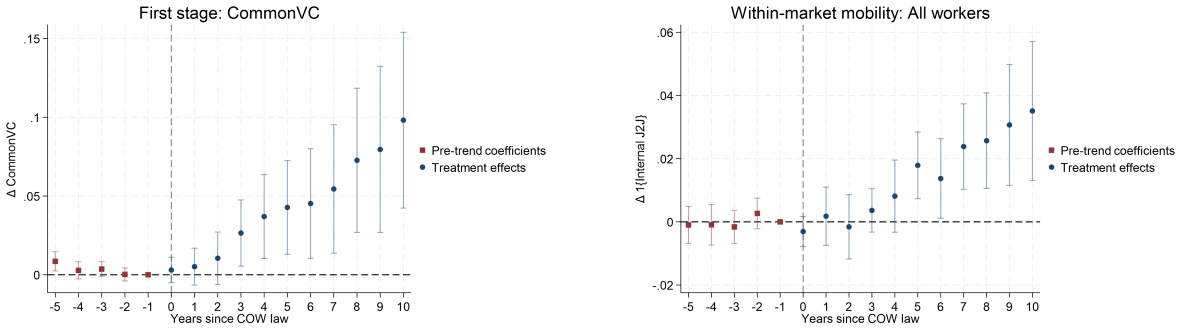
$$Y_{j,t,g} = \sum_{\substack{k=-5 \\ k\neq -1}}^{10} \beta_k \mathbf{1}\{t-g=k\} \cdot D_{j,g} + \mu_{j,g} + \alpha_{c,t} + \gamma_{n,t} + \delta_{g,t} + \varepsilon_{j,t,g}, \tag{2}$$

where $D_{j,g}$ equals one for firms incorporated in the state adopting a COW statute in cohort g. The omitted period is $k = -1$. Because never-treated firms can appear in multiple cohort-specific stacks, we include the fixed effect $\mu_{j,g}$, which allow each firm to have a separate baseline level in each adoption cohort stack. The CBSA-year fixed effects, $\alpha_{c,t}$, absorb time-varying local labor-market shocks, $\gamma_{n,t}$ absorbs industry-year shocks, and $\delta_{g,t}$ absorbs shocks common to all firms in a given cohort stack and calendar year. Standard errors are clustered at the state of incorporation level. This design compares firms incorporated in states that adopt COW statutes to similar firms incorporated in states that never amended their corporate laws to allow for COWs, before and after the institutional change.

The left panel of Figure 2 reports event-study coefficients from Equation (2) using the common VC indicator as the outcome. The pre-treatment coefficients are small and statistically indistinguishable from zero, suggesting that treated and control firms followed similar trends before COW adoption. After adoption, common VC ownership rises steadily, consistent with the interpretation that COW statutes reduce the legal costs of overlapping VC investments. The right panel of Figure 2 report the same specification using job-to-job mobility as the outcome. After COW adoption, worker mobility increase gradually and persistently. The absence of detectable pre-trends alleviates concerns that firms incorporated in adopting states were already experiencing different labor-market trajectories before COW adoption. The timing of the effects is also informative. The gradual post-adoption response is consistent with mechanisms that take time to operate, including the formation of new ownership links and the subsequent transmission of information across portfolio firms.

Panel A of Table 3 sumamrizes the average effects in the 10 years after the adoption.

FIGURE 2. COW adoption, common ownership and worker mobility



A. *CommonVC* B. 𝟙{Internal J2J}, all workers

Notes: Event-study coefficients of the stacked COW design with the post-adoption window extended to ten years: for each adoption cohort g (Delaware 2000, Oklahoma 2001, Missouri 2003, Kansas 2005, Texas 2006, Nevada 2007, New Jersey 2011, Maryland 2014, Washington 2016), firms incorporated in the adopting state and firms incorporated in states that never adopt a corporate opportunity waiver (COW) statute, in years $g-5$ to $g+10$; the cohort data sets are stacked. The dependent variable is regressed on indicators for treated firms in each event year $k = t-g$, with cohort-by-firm, cohort-by-year, industry-by-year and CBSA-by-year fixed effects and no controls. The omitted category is the year before adoption ($k = -1$); squares mark the years before adoption and circles the years from adoption on. Vertical bars are 95% confidence intervals; standard errors clustered at the state-of-incorporation level.

Column 2 shows that, on average, COW adoption increases the annual mobility incidence by 1.4 percentage points, corresponding to a 39.44% increase relative to the estimation-sample mean. These effect are particularly pronounced for C-suite executives (column 4).

3.3. Instrumental variables specification

Finally, we follow Eldar and Grennan (2024) and use eligibility for a Corporate Opportunity Waiver as an instrument for common VC ownership. Let $Z_{j,s,t}$ be an indicator equal to one if firm j is incorporated in state s and that state has adopted a COW statute by year t. For a firm headquartered in state h and operating in industry n, the first-stage equation is

$$CommonVC_{j,s,h,n,t} = \pi Z_{j,s,t} + \alpha_h + \gamma_{n,t} + X'_{j,t}\Lambda_1 + u_{j,s,h,n,t}, \tag{3}$$

where α_h denotes headquarters-state fixed effects, $\gamma_{n,t}$ denotes industry-by-year fixed effects, and $X_{j,t}$ contains the baseline controls for investor characteristics.

For each labor-market outcome $Y_{j,s,h,n,t}$, the corresponding second-stage equation is

$$Y_{j,s,h,n,t} = \beta \widehat{CommonVC}_{j,s,h,n,t} + \delta_h + \eta_{n,t} + X'_{j,t}\Lambda_2 + \varepsilon_{j,s,h,n,t}, \tag{4}$$

TABLE 3. COW adoption, common ownership and within-market mobility: event studies and instrumental variables

		𝟙{Internal J2J}					
	(1) CommonVC	(2) All workers	(3) Innovators	(4) C-suite	(5) Junior	(6) Mid-seniority	(7) Senior
Panel A: COW adoption, stacked design (event-study averages)							
Average of $k = 0, \ldots, 10$	0.0432***	0.0142**	0.00158*	0.00107***	0.00549	0.00626***	0.000965
	(0.0158)	(0.00595)	(0.000924)	(0.000332)	(0.00438)	(0.00215)	(0.000753)
Average of $k = 0, \ldots, 5$	0.0208**	0.00447	-0.00107	0.000323	0.0000671	0.00290*	0.000127
	(0.00948)	(0.00421)	(0.000822)	(0.000231)	(0.00338)	(0.00158)	(0.000561)
Average of $k = 5, \ldots, 10$	0.0655***	0.0245***	0.00440***	0.00194***	0.0112**	0.00991***	0.00167*
	(0.0220)	(0.00770)	(0.00127)	(0.000483)	(0.00541)	(0.00285)	(0.000982)
Observations	2,289,795	2,289,795	2,289,795	2,289,795	2,289,795	2,289,795	2,289,795
Mean DV	0.111	0.0360	0.00561	0.000963	0.0212	0.00993	0.00148
Treated pre-adoption mean	0.0657	0.0247	0.00383	0.000439	0.0139	0.00664	0.000659
Panel B: Instrumental variables							
COW × *Post*	0.105***						
	(0.0100)						
CommonVC		0.183***	0.0533***	0.0102***	0.0926***	0.0797***	0.0145***
		(0.0131)	(0.00791)	(0.00101)	(0.00836)	(0.00630)	(0.00131)
Observations	862,869	862,869	862,869	862,869	862,869	862,869	862,869
Adj. R^2	0.302						
KP Wald F	108.68	108.68	108.68	108.68	108.68	108.68	108.68
Anderson–Rubin p-value		<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Mean DV	0.194	0.0515	0.0108	0.00177	0.0300	0.0165	0.00269
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HQ State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
VC Investor controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Panel A: stacked COW design; for each adoption cohort g, firms incorporated in the adopting state and firms incorporated in never-adopting states, in years $g - 5$ to $g + 10$. Event-study coefficients for $k = t - g$, relative to $k = -1$, with cohort-by-firm, cohort-by-year, industry-by-year and CBSA-by-year fixed effects and no controls; the panel reports their averages over the years shown. Panel B: firm-year panel; column (1) is the first stage of *CommonVC* on *COW* × *Post*, the indicator for COW adoption by the state of incorporation by year t; columns (2)–(7) are 2SLS estimates; industry-by-year and headquarter-state fixed effects and investor controls. KP Wald F is the Kleibergen–Paap F-statistic. Standard errors clustered at the state-of-incorporation level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where $\widehat{CommonVC}_{j,s,h,n,t}$ is the predicted value from Equation (3). Thus, both stages include headquarters-state and industry-by-year fixed effects, as well as the same set of investor controls. We estimate Equations (3) and (4) by two-stage least squares and cluster standard errors at the state-of-incorporation level.

The relevance condition requires COW eligibility to affect the probability of common VC ownership. The left panel in Figure 2 provides strong support for this condition by showing that COW eligibility increases common VC ownership.

The exclusion restriction requires that, conditional on headquarters-state fixed effects,

industry-by-year fixed effects, and investor controls, COW eligibility affects employment and worker mobility only through its effect on common VC ownership. This condition would be violated if COW adoption directly changed firms' financing, governance, or personnel decisions independently of common ownership, or if firms incorporated in adopting states experienced unobserved changes in labor-market outcomes that are not captured by the included fixed effects and controls. We rely on the institutional evidence in Eldar and Grennan (2024), which shows that the statutes were narrowly directed at fiduciary-duty liability, were not adopted in response to broader economic conditions or lobbying by affected firms, and did not systematically alter ventures' incorporation choices.

Additionally, Appendix Figure A.1 and Table A.1 report a placebo analysis examining whether COW adoption affects within-market mobility among ventures without common VC ownership. We repeat the stacked event-study design, restricting treated ventures to those that are never commonly owned during the observed panel or, alternatively, remain without common ownership throughout their cohort's event window, while retaining the full control group. In contrast to the increase observed in the full sample, neither placebo sample exhibits a statistically significant post-adoption increase in mobility. The average effects over event years zero through ten are –0.225 percentage points ($p = 0.601$) and –0.145 percentage points ($p = 0.714$), respectively. These findings are consistent with common ownership being the channel through which COW adoption increases within-market mobility. Lastly, the absence of differential pre-trends in the first-stage event study discussed in Section 3.2 supports the timing and relevance of the instrument, but does not by itself establish the exclusion restriction. Nevertheless, we cannot complitely rule out the existence of smaller direct effects. We therefore interpret the IV estimates as complementary to the stacked difference-in-differences evidence.

Panel B in Table 3 reports instrumental variable (IV) estimates using COW adoption as an instrument for common VC ownership. The first stage is strong: COW adoption increases common VC ownership by 10.5 percentage points, with a KP Wald F-statistic of 108.7. The second-stage estimates are positive and statistically significant across all outcomes, reinforcing our main conclusions.

The IV estimates are larger than the corresponding OLS estimates.[14] We interpret these magnitudes with caution for two reasons. First, the non-instrumented estimates may be downward biased if omitted factors are negatively correlated with common VC

[14]It is not uncommon for IV estimates to be larger than OLS estimates. For example, Samila and Sorenson (2011) estimates the effect of VC on total MSA-level employment and finds IV estimates around five times larger than the corresponding OLS estimates.

ownership but positively correlated with firm-level employment growth and worker mobility.[15] Second, under the usual local average treatment effect interpretation, the IV estimates capture the effect for firms whose likelihood of becoming commonly owned is shifted by COW adoption. These compliers are likely to be firms for which actual and prospective investors' litigation risk previously constrained overlapping ownership or portfolio management activities. The larger IV estimates can therefore also reflect treatment effect heterogeneity: common ownership may have especially large effects when legal frictions were binding and the reform meaningfully expanded investors' ability to shape the governance of portfolio firms.

3.4. The role of governance and local market structure

We next examine whether the effects of common VC ownership are stronger when shared investors have greater ability to influence portfolio-firm decisions. If firms act to maximize shareholder profits, commonly owned firms can be shown to place positive weight on the profits of other firms in the investor's portfolio (Rotemberg 1984; O'brien and Salop 1999). This weight depends on both cash-flow rights and control rights.

Although these weights are not directly observable, board representation provides a natural proxy for investor influence. Board seats give investors greater capacity to monitor managers, advise founders, transmit information across portfolio firms, and shape hiring and personnel decisions. We therefore decompose common VC ownership according to whether a shared investor holds a board seat at the focal firm. Specifically, we construct two mutually exclusive indicators. *CommonVC witBoard Seat* equals one when the focal firm shares a VC investor with another venture in the same industry–CBSA and at least one shared investor holds a board seat at the focal firm. *CommonVC without Board Seat* equals one when the focal firm shares a VC investor with another venture in the same industry–CBSA but no shared investor holds a board seat at the focal firm. Then, we estimate:

$$
\begin{aligned}
1\{J2J\}_{j,n,c,t} = \mu_j + \alpha_{c,t} + \gamma_{n,t} + \beta \cdot \textit{CommonVC with Board Seat}_{j,t} + \\
+ \gamma \cdot \textit{CommonVC without Board Seat}_{j,t} + \mathbf{X}'_{j,t}\Lambda + \varepsilon_{j,n,c,t}.
\end{aligned} \tag{5}
$$

The top panel of Figure 3 illustrates the estimates for β and γ. Common ownership is associated with higher mobility even in the absence of formal board representation, but

[15] For example, safer or more obviously scalable ventures may be more likely to attract capital without relying on overlapping investors, while common ownership may arise precisely when investor coordination, monitoring, or legal constraints are more salient.

the magnitude is four times larger when the shared investor sits on the board. We also examine how mobility evolves around the first common-VC tie, distinguishing firms according to whether a shared investor holds a board seat at the focal firm in the onset year. We estimate separate event-time coefficients for the two groups within a single regression, using never-commonly-owned firms as controls and omitting the year before onset. The specification includes firm, industry-by-year, and CBSA-by-year fixed effects and investor controls. The bottom-left panel of Figure 3 shows that internal job-to-job mobility increases after onset in both groups. From the following year onward, however, the estimated increase is substantially larger among firms with board representation, and the gap widens over subsequent years. These dynamics are consistent with shared investors having greater scope to facilitate worker reallocation when they hold formal control rights.

Next, we examine the role of local market structure, a key determinant of labor-market power. Competition from alternative employers may limit the extent to which commonly owned firms can reduce restrict worker mobility. If greater concentration weakens this constraint, the positive employment and mobility effects of common ownership should attenuate in more concentrated local markets. We examine this possibility by computing an industry-CBSA HHI using employment shares in the year before common ownership first appears in the local market:

$$BaseHHI_{n,c} = \sum_{j} s^2_{j,\tau-1},$$

where τ is the first year in which industry-CBSA (n, c) exhibits any common VC ownership, and $s_{j,\tau-1}$ is firm j's share of employment among ventures in our sample in that industry-CBSA in year $\tau - 1$.[16] We hold this baseline measure fixed over time. Figure C.3 displays its distribution. Then, we estimate:

$$\begin{aligned} 1\{J2J\}_{j,n,c,t} = \mu_j + \alpha_{c,t} + \gamma_{n,t} + \beta \cdot \text{CommonVC}_{j,t} + \\ + \gamma \cdot \Big(\text{CommonVC}_{j,t} \times BaseHHI_{n,c}\Big) + \mathbf{X}'_{j,t}\Lambda + \varepsilon_{j,n,c,t}. \end{aligned} \tag{6}$$

The results in the top panel of Figure 3 show that the positive association between common ownership and job-to-job mobility weakens as local employer concentration increases (the interaction coefficient γ is negative and statistically significant). We complement this specification with event studies that distinguish firms in markets with above-

[16] For cells not observed in $\tau - 1$, we measure *BaseHHI* in τ.

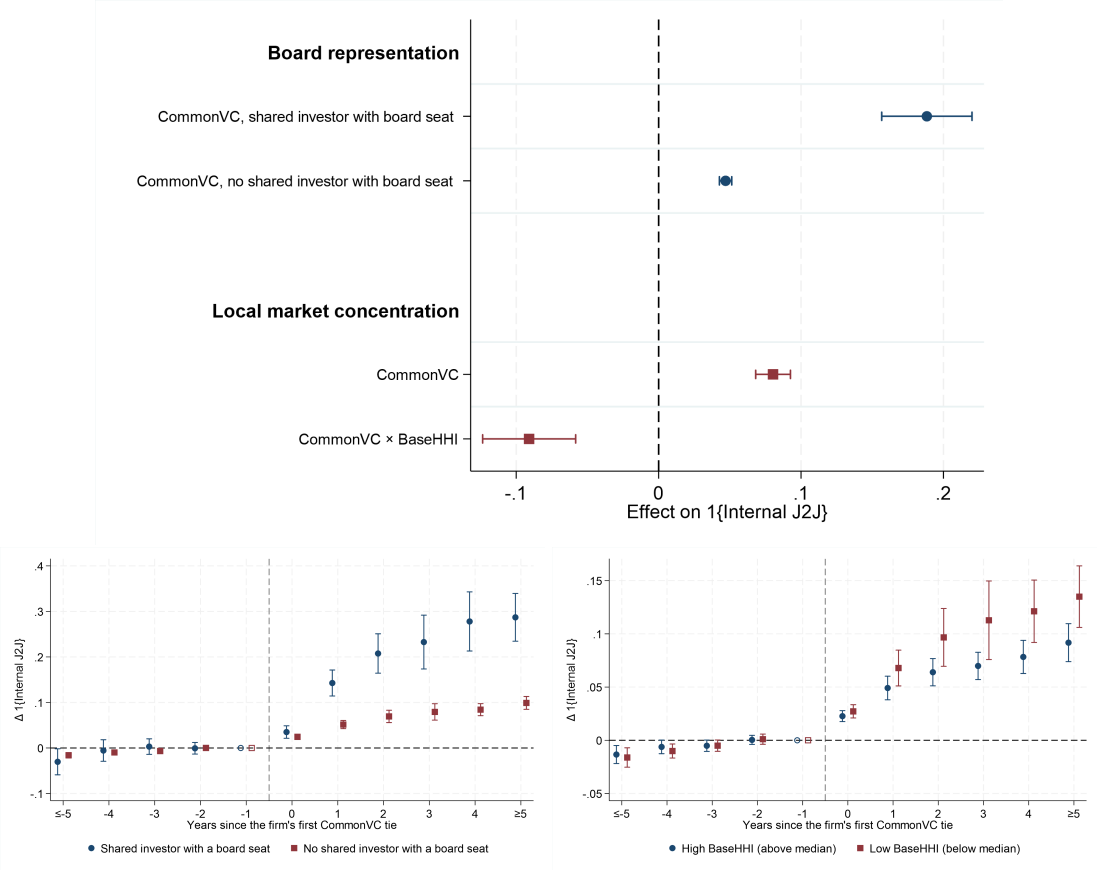


FIGURE 3. Board representation and local market concentration

Notes: The top panel displays the coefficients and 95% confidence intervals for β and γ in Equations (5) and (6). The bottom panel shows event-study coefficients around the year the firm first shares a VC investor with another venture in its industry and CBSA ($k = -1$ omitted; ends binned at $k \leq -5$ and $k \geq 5$), separately for two groups of treated firms in one regression: whether a shared investor holds a board seat at the firm in the onset year (left), and whether *BaseHHI* of the firm's market is above or below the median among treated firms (bottom). Firms never commonly owned are the controls. Firm, industry-by-year and CBSA-by-year fixed effects and investor controls; 95% confidence intervals with standard errors clustered at the industry-CBSA level.

and below-median *BaseHHI*, measured among treated firms. We estimate both groups' event-time coefficients jointly, using the same controls and fixed effects as in the board-representation analysis. The bottom-right panel of Figure 3 shows positive post-onset coefficients in both groups, but the increase is larger and grows more rapidly in less concentrated markets.

These patterns are consistent with common investors continuing to reduce search and screening frictions even in more concentrated markets. Although our HHI measure does not capture all potential employers in the broader industry-CBSA, it is likely to be

informative about concentration within the startup labor market if workers attracted to innovative, VC-backed firms perceive these employers as distinct from larger incumbents or non-VC-backed firms (Bar-Isaac et al. 2025).[17]

4. Internal Labor Markets across Firm Boundaries

Common ownership can create governance and information links among independent firms. We argue that these links can generate an internal labor market across firm boundaries: a setting in which shared investors facilitate matching between workers and jobs across portfolio companies without integrating the firms under common majority control. Prior research on interorganizational networks shows that such ties foster trust, repeated interaction, and information exchange (Gulati 1998; Ahuja 2000). A related literature in labor economics and sociology documents that social networks improve screening and match quality by expanding access to candidates and transmitting credible information about worker ability (Granovetter 1973; Montgomery 1991, 1992).[18]

Venture capital investors are well-positioned to play a similar role across portfolio firms because they often possess detailed information about their investees, their personnel, and their hiring needs. When multiple ventures share the same investor, this informational advantage can reduce uncertainty about prospective matches and facilitate the redeployment of workers to firms where their skills are more valuable. On the worker side, entrepreneurial firms are difficult to evaluate, and workers may have limited information about job quality and growth prospects (Bryan et al. 2026). On the employer side, common ownership can reduce screening and hiring frictions by making information about worker capabilities more visible within the portfolio network. Importantly, these mechanisms need not rely on direct control over hiring decisions. Instead, they operate through increased visibility of job opportunities and worker capabilities via shared board members, advisors, and overlapping professional networks.

To test this mechanism, we construct a dataset of venture pairs located in the same CBSA and estimate the likelihood that at least one worker moves between them in a given year as a function of whether the pair shares a common VC investor. Specifically,

[17] Consistently, in our panel, 5.52% of venture–year observations record at least one worker moving to another VC-backed venture within the same CBSA, compared with 0.85% recording a move to a public firm within the same CBSA.

[18] Consistent with this view, empirical evidence shows that referrals give firms access to otherwise untapped worker pools and mitigate information asymmetries about candidate quality (Beaman and Magruder 2012; Brown et al. 2016; Dustmann et al. 2016; Hensvik and Skans 2016; Pallais and Sands 2016).

we use OLS to estimate the following regression:

$$1\{Mobility\}_{i,j,t} = \alpha_{i,j} + \alpha_t + \beta \cdot SharedVC_{i,j,t} + \\ + \gamma \cdot (SharedVC_{i,j,t} \times SameInd_{i,j}) + \varepsilon_{i,j,t} \tag{7}$$

where $\mathbb{1}\{Mobility\}_{i,j,t}$ is a binary variable equal to one if at least one worker moves between firms i and j, in either direction, during year t; $SharedVC_{i,j,t}$ equals one if the two firms share an investor at t, while $SameInd_{i,j}$ equals one if they operate in the same industry. We also include dyad and year fixed effects, thereby absorbing yearly shocks as well as all persistent pair characteristics such as industry similarity, geographic proximity, or unobserved relational ties. Standard errors are clustered at the CBSA level.[19]

The results in column 1 of Table 4 show that two ventures are significantly more likely to exchange workers when they share a common investor. This effect is larger when the two ventures operate in the same industry, where worker skills are likely to be more transferable (column 2). We then examine whether this pattern is concentrated among workers for whom information frictions are likely to be especially important. Specifically, we repeat the analysis for mobility events involving innovation workers—scientists and engineers—and C-suite executives. These workers are likely to possess more specialized, firm-specific, or strategically valuable knowledge, making external hiring more informationally sensitive. They are also the positions over which VC investors are most likely to have relevant information. Columns 3–6 of Table 4 show that sharing a VC increases the dyad-year probability of mobility for both innovation workers and C-suite executives, with stronger effects for within-industry dyads.[20]

We next examine how bilateral worker flows evolve around the formation of investor connections. We estimate a stacked event study around the first year in which two ventures share a VC investor. For each connection cohort g between 1995 and 2016, we compare pairs first connected in g with pairs first connected after $g+3$, using observations from $g-3$ through $g+3$. Both groups therefore eventually form a shared-investor connection, making the comparison one of connection timing among pairs that develop the same type of relationship. Event-time indicators are interacted with treated-pair status, with the year before connection omitted.

The specification includes pair-by-cohort fixed effects and separate firm-by-year-by-

[19] As a robustness check, we replace the dyad fixed effects with origin-by-year and destination-by-year fixed effects. This addresses the concern that the result is driven by firm-year shocks correlated with the shared-ownership tie rather than by the shared tie itself; the trade-off is that this specification no longer absorbs time-invariant pair match quality. The estimates are robust (Table C.3).

[20] Table C.4 provides additional evidence on within-portfolio worker mobility by job role.

TABLE 4. Within-portfolio worker mobility

	$\mathbb{1}\{\text{Mobility}\} \times 10^4$					
	(1) All workers	(2) All workers	(3) Innovators	(4) Innovators	(5) C-suites	(6) C-suites
SharedVC	16.818***	11.512***	3.409***	2.498***	0.448***	0.286***
	(1.323)	(0.973)	(0.234)	(0.218)	(0.109)	(0.054)
Multiplier	7.7×	5.3×	10.7×	7.8×	12.0×	7.7×
SameInd × *SharedVC*		19.452***		3.336***		0.591**
		(4.440)		(0.716)		(0.266)
Multiplier		8.9×		10.4×		15.8×
Observations	903,782,148	903,782,148	903,782,148	903,782,148	903,782,148	903,782,148
Mean DV	2.184	2.184	0.320	0.320	0.037	0.037
Dyad FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is a directed dyad-year (o, d, t): an ordered pair of distinct ventures, origin o and destination d, in the same CBSA in year t (1990–2021). $\mathbb{1}\{\text{Mobility}\}$ equals one if at least one worker's main job moves from o to d in year t (the main job at o ends in a month of t and the job at d becomes the worker's main job within 12 months; monthly main job built from the Revelio positions), and zero otherwise. Columns (3)–(4) count only movers with scientist or engineer titles and columns (5)–(6) only movers with C-suite titles (chief officers and general counsel), both at the end of the main job at o and at the start of the main job at d. The dependent variable is multiplied by 10^4, so coefficients, standard errors and Mean DV are multiplied by 10^4. *SharedVC* equals one if o and d share a VC investor in year t; *SameInd* equals one if they operate in the same industry. *Multiplier*: coefficient divided by the mean of the dependent variable in the estimation sample (it does not depend on the scaling). All regressions include dyad and year fixed effects; 15,180,488 singleton observations are dropped. Standard errors clustered by CBSA (445 clusters) in parentheses; p-values from t with 444 degrees of freedom. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cohort fixed effects for both ventures. Pair effects absorb persistent bilateral characteristics, including geographic proximity, industry similarity, and preexisting relational ties. Firm-by-year effects absorb annual conditions at each venture, including general changes in hiring, departures, and firm activity. The estimates thus capture changes in worker exchange specific to the newly connected pair, conditional on persistent pair characteristics and annual conditions at both firms. Identification requires that, absent the connection, bilateral mobility among newly connected pairs would have evolved similarly to that of later-connected pairs, conditional on these fixed effects. Standard errors are clustered at the CBSA level.

Figure 4 shows that the probability of worker exchange rises following the first shared investor. The average coefficient over event years zero through three corresponds to an increase of 0.132 percentage points. The two preconnection coefficients are not jointly statistically distinguishable from zero.

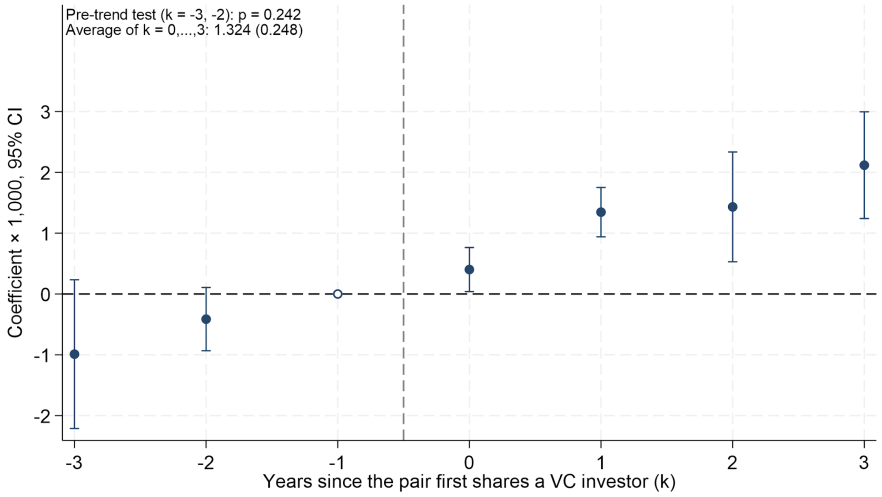
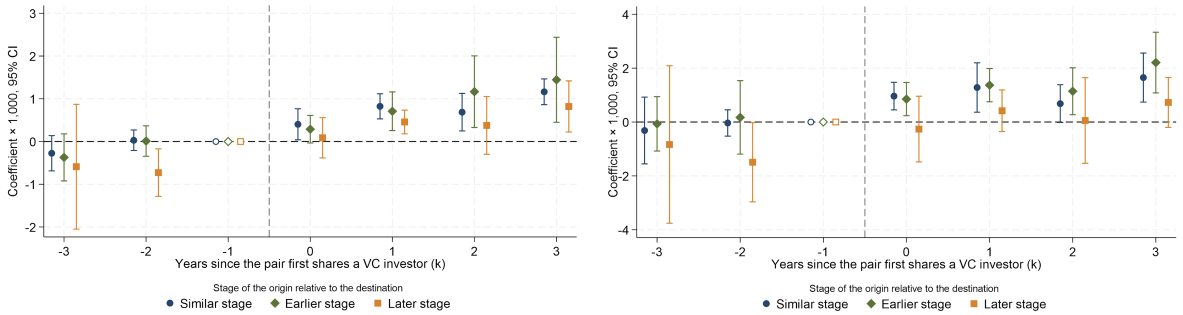


FIGURE 4. Job-to-job flows between venture pairs around the first shared VC investor

Notes: Event-study coefficients ($\times$1,000) for any job-to-job flow between two ventures in the same CBSA, around the first year in which the pair shares a VC investor ($k = 0$); $k = -1$ is omitted. Stacked cohorts with not-yet-connected pairs as controls. Cohort-by-dyad, cohort-by-origin-by-year and cohort-by-destination-by-year fixed effects. 95% confidence intervals with standard errors clustered by CBSA.

An internal labor market can connect ventures with different organizational needs as they develop. A venture building its team may benefit from experienced workers leaving a company further along its development path, while a more advanced venture may offer opportunities to workers departing an earlier-stage firm. Shared investors may facilitate both types of movement by observing staffing needs and worker capabilities across their portfolios. The direction of worker flows across financing stages therefore helps characterize how investor networks contribute to the allocation of workers: whether connected mobility primarily involves ventures at similar stages of development or also links firms with potentially different personnel needs. We examine this question using completed VC financing rounds as an observable measure of relative development. We classify directed pairs using the difference in completed VC financing rounds in the year the pair first shares an investor. Ventures are classified as being at the same stage when their numbers of completed rounds differ by no more than one. A flow runs from a more advanced to a less advanced venture when the origin has completed at least two more rounds than the destination; the reverse ordering defines flows from less advanced to more advanced ventures. We estimate separate event-time responses for the three groups, retaining directed-pair-by-cohort, origin-by-year-by-cohort, and destination-by-year-by-cohort fixed effects. We additionally include stage-group-by-year-by-cohort fixed effects, allowing each relative-stage category its own background time path within each stack.



A. Any job-to-job

B. Any within-industry job-to-job

FIGURE 5. Worker flows between venture pairs and differences in financing stage

Notes: Event-study coefficients (×1,000) of directed pair flows around the first shared VC investor, separately by the stage of the origin relative to the destination (similar stage: VC rounds within one of each other; earlier or later stage: at least two rounds apart). The left (right) panel considers ventures within the origin CBSA (industry-CBSA) as potential destinations. Cohort-by-dyad, cohort-by-origin-by-year and cohort-by-destination-by-year fixed effects; 95% confidence intervals with standard errors clustered by CBSA.

Figure 5 shows that worker exchange following a shared-investor connection extends both across and within financing stages. The probability of a move increases between similarly staged ventures and from earlier-stage origins to later-stage destinations, with the largest estimated increases appearing toward the end of the post-connection window. These patterns are present both when destinations include ventures across industries within the origin's CBSA and when destinations are restricted to the same industry and CBSA. Evidence for flows from later-stage origins to earlier-stage destinations is weaker, particularly within the same industry, where post-connection estimates are small and imprecise. The findings are therefore consistent with investor networks connecting workers to opportunities across financing stages, with clearer cross-stage evidence for movement toward ventures that have completed more funding rounds.

4.1. Common VC ownership and worker redeployment

Having shown that worker flows follow investor connections, we next examine whether shared investors facilitate reemployment within the portfolio network by studying where separating workers find their next job. Unlike the dyadic analysis, which records whether any worker moves between two firms in either direction, the destination analysis follows individual separations and identifies the firm that hires each worker.

We consider workers separating from an active venture, and for each origin i and separation year t, the set of potential destinations, $\mathcal{J}_{i,t}$, contains every other venture active in the origin's CBSA in that year, across industries. Each worker is paired with

every firm in this set. The outcome $D_{w,i,j,t}$ equals one if worker w's next job begins at destination j within 36 months of separation and zero otherwise. Workers with no observed next job within that horizon, or whose next job is outside the choice set, have zero outcomes for every potential destination. We estimate

$$D_{w,i,j,t} = \alpha_{i,t} + \delta_j + \beta\, SharedVC_{i,j,t} + \gamma\, SameInd_{i,j} + \varepsilon_{w,i,j,t}. \tag{8}$$

Here, $SharedVC_{i,j,t}$ indicates that the origin and potential destination share an investor in the separation year. Origin-by-year fixed effects absorb conditions common to workers leaving the same firm in the same year, and destination fixed effects absorb persistent differences in firms' attractiveness and hiring activity. We also estimate specifications that interact *SharedVC* with *SameInd*. Standard errors are clustered at the CBSA level.[21]

Table 5 distinguishes two samples. The preferred specifications in Panel A, columns (1)–(2), include all separations for which the full 36-month follow-up window is observed. Panel B conditions on workers whose next job is at a venture in the choice set within 36 months. Each worker in Panel B therefore has exactly one chosen destination, allowing us to examine how investor connections relate to the allocation of locally reemployed workers across potential employers.

Panel A, column (1), shows that a shared investor is associated with a higher probability that a separating worker joins a given potential destination. The coefficient corresponds to an increase of 0.0148 percentage points, approximately 6.4 times the pooled mean probability across worker–potential destination observations. The association is larger for destinations in the origin's industry: the positive interaction in column (2) indicates that shared investor connections are particularly relevant where workers' skills may be more transferable.

Panel B shows that investor connections also predict destination choice among workers reemployed within the local venture market. The coefficient in column (1) corresponds to an increase of 0.118 percentage points in the probability of choosing a particular destination, relatve to a mean of 0.052 percent. The same-industry interaction remains positive in column (2), and the connected-destination association remains statistically significant when destination-by-year fixed effects absorb annual hiring conditions at each potential employer (column 3). Taken together, the two panels show that connected firms disproportionately receive departing workers, both in the full separa-

[21] Because the regressors are constant within each origin–destination–year cell, we estimate the model using cell-average outcomes weighted by the number of workers represented in the cell; this produces the same slope coefficients as the worker-level regression.

TABLE 5. Shared investors and the destination of separating workers

VARIABLES	(1) $\mathbb{1}\{\text{move to } j\}$	(2) $\mathbb{1}\{\text{move to } j\}$	(3) $\mathbb{1}\{\text{move to } j\}$	(4) $\mathbb{1}\{\text{move to } j\}$
Panel A: All separations, every venture in the origin's CBSA as a potential destination				
SharedVC	1.481***	0.913***	1.695***	0.768***
	(0.346)	(0.206)	(0.400)	(0.212)
SameInd	0.737***	0.684***	0.882***	0.540***
	(0.110)	(0.103)	(0.133)	(0.089)
SharedVC × *SameInd*		2.401***		
		(0.683)		
Observations	9,872,740,625	9,872,740,625	8,081,928,540	12,897,619,797
Mean DV (×10,000)	0.233	0.233	0.284	0.178
Multiplier	6.4×	3.9×	6.0×	4.3×
Sample	Observed 36m	Observed 36m	Reemployed in 36m	Incl. censored
Origin-Year FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Panel B: Workers reemployed at a venture in the choice set within 36 months				
	(1)	(2)	(3)	(4)
SharedVC	11.834***	8.376***	10.876***	11.774***
	(4.149)	(2.764)	(3.919)	(4.130)
SameInd	11.896***	11.415***	11.596***	11.878***
	(2.888)	(2.782)	(2.807)	(2.882)
SharedVC × *SameInd*		12.178**		
		(5.546)		
Observations	442,023,962	442,023,962	441,997,786	442,023,966
Mean DV (×10,000)	5.195	5.195	5.183	5.195
Multiplier	2.3×	1.6×	2.1×	2.3×
Origin-Year FE	Yes	Yes	Yes	No
Destination FE	Yes	Yes	No	Yes
Destination-Year FE	No	No	Yes	No
Origin FE	No	No	No	Yes
Year FE	No	No	No	Yes

Notes: The unit of observation is a worker–origin firm–potential destination firm–year-of-separation combination (w, i, j, t), where the potential destinations are a worker-specific choice set consisting of all VC-backed ventures active in the origin firm's CBSA in the year of separation, in any industry. The dependent variable equals one if the worker moves from the origin to that destination within 36 months and zero otherwise; coefficients, standard errors and means are multiplied by 10,000. Estimation uses the origin–destination–year cell means weighted by the number of separations. Panel A uses all separations from active ventures with the 36-month window observed (columns 1–2), separations followed by a job within 36 months (column 3), and all separations including those whose window is right-censored (column 4). Panel B conditions on workers reemployed at a venture in the choice set within 36 months, so that each worker has exactly one realized destination and the estimates describe the destination-choice margin. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

tion sample and conditional on reemployment in the local choice set.[22] The contrast between the two panels is informative: because Panel A assigns zeros to workers with no observed subsequent job, its estimate combines a higher probability of reemployment within the local venture set with the direction of realized moves, whereas Panel B isolates the allocation margin alone. The Panel A multiplier substantially exceeds that of Panel B, indicating that investor connections not only direct locally reemployed workers toward connected firms but also raise the likelihood that a separating worker is reemployed within the local venture market at all.

Separate evidence on the reemployment margin is reported in Panel A of Appendix Table C.8. Common VC ownership at the origin is positively associated with the probability that a separating worker's next job is at a venture in the local choice set within 36 months, for both small and large origin firms. The corresponding estimates for reemployment at any employer are small and statistically insignificant.

Panel B of Appendix Table C.8 examines whether the connected-destination association varies with the origin firm's size among locally reemployed workers. The association is larger for workers leaving smaller ventures, using either employment or cumulative capital raised to measure size. The differences are statistically significant at the 5% level for employment and the 10% level for capital raised. This pattern is consistent with an informational role for common investors: individual workers' capabilities may be easier to observe in smaller organizations. Size also captures differences in organizational structure and hiring needs, so the comparison supports this interpretation without isolating information transmission as the sole explanation.

4.2. The speed of connected transitions

We next consider the speed of the transition among workers whose next job begins within 36 months at a venture in the origin's CBSA choice set. Here, each observation is a worker's realized origin–destination transition, and *SharedVC* describes the connection between the origin and the firm that actually hires the worker. We estimate

$$T_{w,i,j,t} = \alpha_{i,t} + \delta_j + \beta\, SharedVC_{i,j,t} + \gamma\, SameInd_{i,j} + \varepsilon_{w,i,j,t}, \tag{9}$$

[22] Appendix analyses in Tables C.6 and C.7 show a positive connected-destination association across horizons from six to 36 months and among scientists and engineers, high-seniority workers, and C-suite executives. The horizon-specific samples differ because follow-up requirements and observed reemployment vary with the horizon.

TABLE 6. Shared investors and transition speed among locally reemployed workers

	(1)	(2)	(3)	(4)
	𝟙{next job in 6m}	𝟙{next job in 6m}	log days to new job	log days to new job
SharedVC	0.007**	0.013*	-0.041*	-0.065
	(0.004)	(0.007)	(0.024)	(0.046)
SameInd	0.017***	0.019***	-0.141***	-0.147***
	(0.004)	(0.004)	(0.021)	(0.016)
SharedVC × *SameInd*		-0.013		0.053
		(0.011)		(0.073)
Observations	178,075	178,075	178,075	178,075
Mean DV	0.818	0.818	2.406	2.406
Origin-Year FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes

Notes: OLS estimates. The unit of observation is a worker–origin firm–realized destination–year of separation. Separations and reemployment are spell-based (v6): a separation is the end of a Revelio employment spell, and the new job is the worker's next spell. The sample consists of workers separating from an active private venture who are reemployed within 36 months at a venture in the origin's CBSA choice set, with the 36-month window observed (it ends by December 2021, so separations up to 2018). *SharedVC* equals one if the origin and the realized destination share a VC investor in the year of separation, and *SameInd* equals one if they operate in the same industry. Log days to new job is $\ln(1 + \max(0, \text{days between separation and the start of the next spell}))$: a next spell that starts before the separation counts as zero days. Singleton fixed-effect groups are dropped. Standard errors clustered at the CBSA level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where $T_{w,i,j,t}$ is either an indicator that the next job begins within six months or $\ln(1 + \textit{days})$, with *days* denoting the interval between separation and the next job's start. As in the destination analysis, we include origin-by-year and destination fixed effects and cluster standard errors at the CBSA level. Columns (2) and (4) of Table 6 additionally interact shared ownership with the same-industry indicator.

Column (1) in Table 6 shows that moving to a destination connected through a shared investor is associated with a 0.7 percentage-point increase in the probability of beginning the new job within six months, relative to a sample mean of 82%. The coefficient in column (3) also indicates faster transitions to connected firms, with an implied reduction in the number of days of approximately 4%, altough this estimate is only significant at the 10% level. Same-industry transitions also occur more quickly. However, the interaction estimates in columns (2) and (4) provide little evidence that the additional speed advantage associated with a shared investor differs systematically between within- and across-industry moves.

4.3. Redeployment around firm contractions

Next, we examine worker redeployment around both adverse developments and successful liquidity events at the origin. We first repeat the destination analysis for workers leaving ventures whose employment declines by at least 50 or 70 percent between the separation year and the following year. Table 7 reports specifications using all separations with complete 36-month follow-up and specifications conditional on reemployment in the local choice set. The connected-destination association is positive and statistically significant in both samples and at both contraction thresholds. Among locally reemployed workers, the estimated increase in the probability of choosing a given connected destination is 0.134 percentage points at the 50 percent threshold and 0.138 percentage points at the 70 percent threshold. Investor connections therefore remain relevant to the allocation of workers leaving sharply contracting ventures. Appendix Table C.10 shows that the positive connected-destination association extends across occupations and seniority groups, with the largest point estimates for C-suite executives and high-seniority workers.

Figure 6 traces the connected-destination association around the origin's first employment decline of at least 50 percent, using the year before contraction as the reference period. The sample includes departures from ventures that eventually experience such a contraction. Among workers reemployed within the local venture choice set, the association is higher in the contraction year and the following year than in the reference year, both for the full worker sample and when we restrict attention to C-suite executives.

We also consider the reallocation of personnel following IPOs. This analysis is motivated by evidence that internal labor markets in business groups facilitate the movement of workers with general management skills as firms' needs change (Huneeus et al. 2021). Workers, and particularly executives, leaving a venture after an IPO may possess experience that is valuable to other entrepreneurial firms seeking managerial talent. We therefore ask whether departing employees disproportionately join ventures connected through the origin's VC investors.

Using the same destination-choice framework, Appendix Table C.11 shows that shared investors also predict the allocation of workers leaving ventures in or after their IPO year. Among workers reemployed within 36 months at a venture in the origin's CBSA, a shared investor is associated with an increase in the probability of choosing a given destination of 0.126 percentage points for all workers and 0.189 percentage points for C-suite executives—with both estimates statistically significant at the one-percent level.

Taken together, the evidence is consistent with shared investors facilitating reemployment within an internal labor market that spans formally independent firms. Work-

TABLE 7. Shared investors and the destination of workers leaving contracting ventures

	(1)	(2)	(3)	(4)
	Decline $\geq$ 50%		Decline $\geq$ 70%	
Sample	All	In choice set	All	In choice set
SharedVC	2.134***	13.407***	2.312***	13.832***
	(0.438)	(3.350)	(0.490)	(3.753)
SameInd	1.096***	10.122***	1.298***	11.158***
	(0.217)	(2.112)	(0.247)	(2.334)
Observations	243,462,790	22,505,643	114,798,411	11,697,495
Mean DV ($\times$10,000)	0.456	4.919	0.514	5.022
Multiplier	4.7$\times$	2.7$\times$	4.5$\times$	2.8$\times$
Origin-Year FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes

Notes: The unit of observation is a worker–origin firm–potential destination firm–year-of-separation combination (w, i, j, t), where the potential destinations are a worker-specific choice set consisting of all VC-backed ventures active in the origin firm's CBSA in the year of separation, in any industry. The dependent variable equals one if the worker moves from the origin to that destination within 36 months and zero otherwise, including all zeros for workers with no subsequent job and for workers whose next job is outside the choice set; coefficients, standard errors and means are multiplied by 10,000. *SharedVC* equals one if the origin and the potential destination share a VC investor in the year of separation, and *SameInd* equals one if they operate in the same industry. Estimation uses the origin–destination–year cell means weighted by the number of separations, which is numerically identical to a worker-level regression. The sample is restricted to workers separating from ventures experiencing an employment decline of at least 50 percent (columns 1–2) or 70 percent (columns 3–4) between the year of separation and the following year. All uses all separations from these ventures with the 36-month window observed; In choice set conditions on workers reemployed at a venture in the choice set within 36 months. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

ers disproportionately move to connected ventures, and the destination relationship is stronger where an informational role for investors is plausible. The persistence of connected flows after contractions and IPOs further shows that these networks remain relevant when an origin sharply reduces employment or after positive shocks.

4.4. Financing decisions and worker redeployment

Shared investors may in principle influence both the matching of workers to portfolio firms and the financing decisions that lead workers to leave their employers. For example, an investor may withdraw support from one venture and redirect its resources

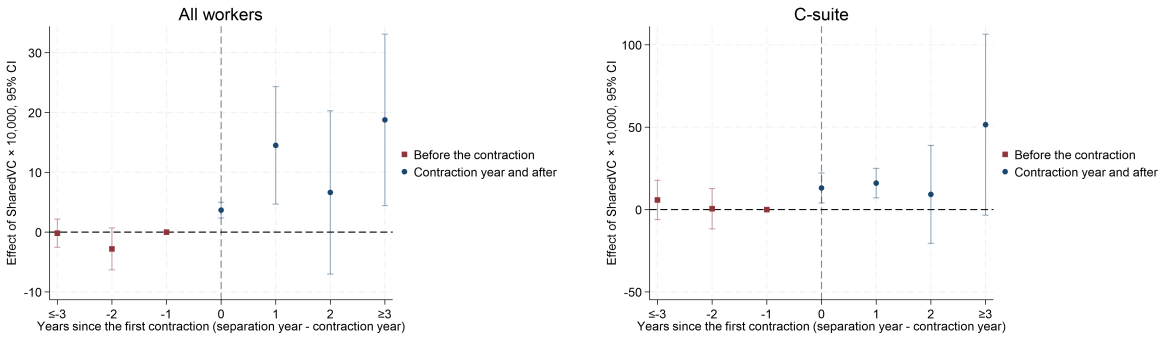


FIGURE 6. Redeployment around contractions

Notes: Event studies of the destination choice of workers who leave a venture, around the venture's first contraction. Ventures with a contraction, defined as an employment decline of at least 50% between year c and the venture's next observed year in the firm-year panel, where c is the first such year. The sample is the workers who separate from these ventures in year $c + k$, with $k \leq -3, \ldots, \geq 3$ and the end points binned, and who are reemployed within 36 months at a VC-backed venture in the origin's CBSA (the choice set; the 36-month window is observed). Each worker is paired with every potential destination in the choice set, and the outcome equals one for the destination the worker moves to. The regression of this indicator, estimated on origin–destination–year cells weighted by the number of reemployed workers, includes *SharedVC* (the origin and the potential destination share a VC investor in the year of separation), *SameInd* (the two ventures are in the same industry) and their interactions with event-time indicators, together with origin-by-year and destination fixed effects. The figure plots the coefficients on $SharedVC \times \mathbb{1}\{k\}$, relative to $k = -1$ and multiplied by 10,000. C-suite panels restrict the sample to workers who held a C-suite title in the origin spell. Every venture in the sample experiences a contraction, so there is no comparison group of ventures without it. Vertical bars are 95% confidence intervals with standard errors clustered by CBSA.

toward another portfolio company (Li et al. 2023; Leccese 2024). This would change the interpretation of increased mobility, given that the investor might help create the separations as well as facilitate subsequent matches. We examine whether the observed financing histories and worker flows are consistent with this account.

We use follow-on investment in the origin as an observable proxy for continued financial support by the shared investor. If connected worker flows reflect investors withdrawing support from one venture and redeploying its workers elsewhere, we would expect a stronger connected-destination association among contracting origins without follow-on investment than among those receiving additional capital. For each connected origin–destination pair, we therefore identify whether a shared investor invested again in the origin after the first year in which it held both ventures and through one year after separation. We replace *SharedVC* in Equation (8) with two separate indicators for connected pairs with and without this additional investment in the origin, retaining origin-by-year and destination fixed effects.

Appendix Table C.12 shows a positive association in both categories. For locally

reemployed workers leaving ventures contracting by at least 50 percent, the coefficients are 0.120 percentage points when a shared investor provides follow-on financing to the origin and 0.139 percentage points when no such investment is observed. Their difference is not statistically significant. The other samples and the 70 percent threshold give the same qualitative pattern. Thus, investor connections predict worker destinations both when follow-on financing to the contracting origin is observed and when it is not.

We provide additional evidence supporting this interpretation by examining financing at both the origin and destination. If shared investors withdraw support from one venture and redeploy its workers elsewhere, we would expect stronger flows from origins without further financing toward destinations receiving additional support. We examine this pattern in an annual panel of directed venture pairs in the same CBSA that have completed their Series A rounds.[23] Among pairs sharing a Series A investor, we distinguish subsequent investment by shared Series A investors in both ventures, only the origin, only the destination, or neither, measured through the year of the flow. Pairs without a shared Series A investor form the reference group. Appendix Table C.13, column (2), shows the largest coefficient when both ventures receive follow-on financing. The coefficient is also larger when only the origin receives financing than when only the destination does. This ordering persists with origin-by-year and destination-by-year fixed effects in column (7).

Overall, although these analyses cannot establish whether common ownership caused individual ventures to contract, the evidence does not support the view that the observed mobility is primarily driven by investors withdrawing support from one venture and redeploying its workers elsewhere in the portfolio.

5. Post-transition Outcomes: Retention and Promotion

The evidence so far shows that workers disproportionately move to connected ventures and that these transitions occur more quickly. How do workers fare after these moves? Faster transitions could reflect improved matching, but could also place workers in positions that are less durable or offer limited career progression. Since information about the suitability of a match can emerge after hiring (Jovanovic 1979), subsequent retention provides one indication of whether connected transitions lead to productive matches. We examine employment-spell duration alongside within-firm promotion, which provides a complementary measure of workers' subsequent outcomes.

[23] A Series A round is an early-stage equity financing round, typically following seed funding and involving venture capital investors, through which a venture raises capital to develop its product and expand its operations.

TABLE 8. Tenure and promotions at connected destinations

VARIABLES	(1) Any separation	(2) Exit to non-employment	(3) Exit to another job	(4) First promotion
SharedVC	-0.021*	-0.027	-0.023*	0.146***
	(0.012)	(0.027)	(0.012)	(0.034)
SameInd	-0.034***	-0.005	-0.041***	-0.095***
	(0.011)	(0.019)	(0.010)	(0.020)
Spells	221,773	221,773	221,773	221,773
Events	147,956	26,986	119,740	20,638
Censored spells	73,817	194,787	102,033	201,135
Implied hazard ratio (*SharedVC*)	0.979	0.973	0.978	1.158
Entry seniority & occupation controls	Yes	Yes	Yes	Yes
Strata: Dest. industry × entry year	Yes	Yes	Yes	Yes

Notes: The unit of observation is an employment spell of a worker at the destination firm: workers separating from an active venture and reemployed within 36 months at a venture in the origin's CBSA choice set, with consecutive positions at the same firm collapsed into a single spell. *SharedVC* equals one if the origin and the destination share a VC investor in the year of the move, and *SameInd* equals one if they operate in the same industry. Columns (1)–(4) report Cox proportional-hazard estimates (Efron method for ties) with the baseline hazard stratified by destination-industry-by-entry-year cohort. Column (1) uses any separation as the failure event; spells are right-censored when the position is ongoing at the end of the sample or when the destination is acquired or undergoes an IPO. Columns (2) and (3) report cause-specific hazards: the failure event is an exit with no new position within twelve months (column 2) or an exit followed by a new position within twelve months (column 3), with the other exit type treated as censoring. Column (4) uses the first year in which the worker's seniority exceeds the entry level as the failure event, with separation treated as censoring. A worker who leaves the destination before being promoted counts as not promoted. A spell ends at the last end date of its positions; exits in the final twelve months of the data without a new position are separations in column (1) and are censored in columns (2) and (3). All specifications control for entry seniority and occupation. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

We construct a spell-level dataset from the realized transitions analyzed in Table 6: workers separating from an active venture and reemployed within 36 months at a venture in the origin's CBSA choice set. Consecutive positions at the destination are collapsed into a single employment spell. We record the spell's start and end dates and the worker's seniority at entry and in subsequent positions.[24] For completed spells, we distinguish exits followed by a recorded new position within twelve months from exits without one.

We estimate Cox proportional-hazards models of the form

$$\lambda_{w,i,j}(\tau) = \lambda_{0,s(j,t)}(\tau)\, \exp\big(\beta\, SharedVC_{i,j,t} + \gamma\, SameInd_{i,j} + \mathbf{X}'_w \Lambda\big), \tag{10}$$

[24]Spells are right-censored at the end of the sample or when the destination firm is acquired or undergoes an IPO, so the analysis concerns employment outcomes before these ownership changes.

where τ denotes tenure at the destination, $SharedVC_{i,j,t}$ indicates that the origin and destination shared a VC investor in the year of the move, $SameInd_{i,j}$ indicates that they operate in the same industry, and $\mathbf{X}_w$ contains entry-seniority and occupation controls. The baseline hazard is stratified by destination industry and entry year, so comparisons are made within industry–entry-year cohorts. Standard errors are clustered at the CBSA level. We estimate the overall separation hazard and cause-specific hazards for the two exit types, treating the competing exit type as censoring. We also estimate the hazard of a first within-firm promotion, defined as the case in which seniority exceeds the entry level, with separation treated as censoring.[25]

Table 8 reports the results. Column (1) shows no statistically significant increase in overall separation hazards: the estimated hazard ratio for *SharedVC* is 0.979. Distinguishing exit types yields similar results.[26] However, we find that workers moving to connected destinations also experience faster promotions. Column (4) implies a 15.8% higher promotion hazard among workers still employed and not yet promoted.

These comparisons are conditional on realized local reemployment and do not identify a causal effect of investor intermediation on worker post-transition outcomes. Nevertheless, the broadly similar retention and higher recorded promotion hazards provide evidence of favorable post-transition outcomes at connected destinations. These patterns are consistent with investor networks facilitating productive matches across firm boundaries.

6. Conclusion

This paper studies how common VC ownership shapes worker mobility. Combining venture-capital investment data with worker employment histories, we find that common ownership is associated with greater job-to-job mobility. COW adoption increases common ownership and is followed by similar increases in employee mobility, with substantially larger changes when shared investors hold board seats. At the same time, the documented positive associations are smaller in more concentrated local markets, suggesting that the consequences of common ownership may depend on market structure.

We further show that worker moves are disproportionately more likely between firms connected through a shared investor, including among workers leaving sharply

[25] This measure captures recorded seniority advancement. The promotion hazard describes advancement among workers who remain employed at the destination and have not yet been promoted. Similarly, the exit-specific hazards describe separation rates among workers still employed there.

[26] Exit classification is incomplete for departures in the final year of the employment histories. Appendix Table C.15 ends employment-spell follow-up in December 2020, censoring ongoing spells at that cutoff and using the 2021 histories to classify departures occurring by then. The results are robust to this check.

contracting ventures. Among workers reemployed locally, transitions to connected destinations happen faster, and the resulting matches show no evidence of worsened worker outcomes, as measured by retention and promotion likelihood. These patterns are consistent with ownership links creating an internal labor market across firm boundaries through active portfolio intermediation.

Overall, our findings suggest that common ownership can affect labor markets not only by changing firms' incentives to compete for workers, but also by shaping the allocation of human capital across firms.

References

Adhvaryu, A., J.-F. Gauthier, A. Nyshadham, and J. Tamayo (2024): "Absenteeism, Productivity, and Relational Contracts Inside the Firm," *Journal of the European Economic Association*, 22, 1628–1677.

Agrawal, A., I. Cockburn, and J. McHale (2006): "Gone but not forgotten: knowledge flows, labor mobility, and enduring social relationships," *Journal of Economic Geography*, 6, 571–591.

Ahuja, G. (2000): "Collaboration networks, structural holes, and innovation: A longitudinal study," *Administrative science quarterly*, 45, 425–455.

Amanzadeh, N., A. Kermani, and T. McQuade (2024): "Return migration and human capital flows," Tech. rep., National Bureau of Economic Research.

Amornsiripanitch, N., P. A. Gompers, and Y. Xuan (2019): "More than money: Venture capitalists on boards," *The Journal of Law, Economics, and Organization*, 35, 513–543.

Antón, M., F. Ederer, M. Giné, and M. Schmalz (2023): "Common ownership, competition, and top management incentives," *Journal of Political Economy*, 131, 1294–1355.

——— (2025): "Innovation: the bright side of common ownership?" *Management Science*, 71, 3713–3733.

Arnold, D. (2019): "Mergers and acquisitions, local labor market concentration, and worker outcomes," *Local Labor Market Concentration, and Worker Outcomes (October 27, 2019)*.

Azar, J., Y. Qiu, and A. Sojourner (2022): "Common ownership in labor markets," Tech. rep., WE Upjohn Institute for Employment Research.

Azar, J., M. C. Schmalz, and I. Tecu (2018): "Anticompetitive effects of common ownership," *The Journal of Finance*, 73, 1513–1565.

Backus, M., C. Conlon, and M. Sinkinson (2021a): "Common ownership and competition in the ready-to-eat cereal industry," Tech. rep., National Bureau of Economic Research.

——— (2021b): "Common Ownership in America: 1980–2017," *American Economic Journal: Microeconomics*, 13, 273–308.

Baker, G. and B. Holmstrom (1995): "Internal labor markets: Too many theories, too few facts," *The American Economic Review*, 85, 255–259.

Balasubramanian, N., J. W. Chang, M. Sakakibara, J. Sivadasan, and E. Starr (2022): "Locked

in? The enforceability of covenants not to compete and the careers of high-tech workers," *Journal of Human Resources*, 57, S349–S396.

Bar-Isaac, H., J. P. Johnson, and V. Nocke (2025): "Acquihiring for Monopsony Power," *Management Science*, 71, 3485–3496.

Barone, G., F. Schivardi, and E. Sette (2025): "Interlocking Directorates and Competition in Banking," *The Journal of Finance*, 80, 1963–2016.

Beaman, L. and J. Magruder (2012): "Who gets the job referral? Evidence from a social networks experiment," *American Economic Review*, 102, 3574–3593.

Belenzon, S. and U. Tsolmon (2016): "Market frictions and the competitive advantage of internal labor markets," *Strategic Management Journal*, 37, 1280–1303.

Bernstein, S., X. Giroud, and R. R. Townsend (2016): "The impact of venture capital monitoring," *The Journal of Finance*, 71, 1591–1622.

Bernstein, S., R. R. Townsend, and T. Xu (2023): "Flight to Safety: How Economic Downturns Affect Talent Flows to Startups," *The Review of Financial Studies*, 37, 837–881.

Brown, M., E. Setren, and G. Topa (2016): "Do informal referrals lead to better matches? Evidence from a firm's employee referral system," *Journal of Labor Economics*, 34, 161–209.

Bryan, K. A., M. Hoffman, and A. Sariri (2026): "Information Frictions and Employee Sorting between Start-ups," *American Economic Journal: Applied Economics*, 18, 369–406.

Burnside, A. J. and A. Kidane (2020): "Common ownership: an EU perspective," *Journal of Antitrust Enforcement*, 8, 456–510.

Callaci, B., M. Gibson, S. Pinto, M. Steinbaum, and M. Walsh (2024): "The effect of franchise no-poaching restrictions on worker earnings," *Review of Economics and Statistics*, 1–35.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019): "The effect of minimum wages on low-wage jobs," *The Quarterly Journal of Economics*, 134, 1405–1454.

Cestone, G., C. Fumagalli, F. Kramarz, and G. Pica (2024): "Exploiting Growth Opportunities: The Role of Internal Labour Markets," *The Review of Economic Studies*, 91, 2676–2716.

Chen, A., M. B. Zhang, and Z. Zhang (2025): "Talent market competition and firm growth," *Available at SSRN 4597388*.

Chen, H., P. Gompers, A. Kovner, and J. Lerner (2010): "Buy local? The geography of venture capital," *Journal of Urban Economics*, 67, 90–102, special Issue: Cities and Entrepreneurship.

Coff, R. W. (1997): "Human assets and management dilemmas: Coping with hazards on the road to resource-based theory," *Academy of Management Review*, 22, 374–402.

Dasaratha, K. (2023): "Innovation and strategic network formation," *The Review of Economic Studies*, 90, 229–260.

De Chaisemartin, C. and X. d'Haultfoeuille (2020): "Two-way fixed effects estimators with heterogeneous treatment effects," *American Economic Review*, 110, 2964–2996.

Delabastita, V. and M. Rubens (2025): "Colluding against workers," *Journal of Political Economy*, 133, 1796–1839.

Dustmann, C., A. Glitz, U. Schönberg, and H. Brücker (2016): "Referral-based job search networks," *The Review of Economic Studies*, 83, 514–546.

Ederer, F. and B. Pellegrino (2026): "A Tale of Two Networks: Common Ownership and Product Market Rivalry," *The Review of Economic Studies*, 93, 1746–1788.

Eldar, O. and J. Grennan (2021): "Common Ownership and Entrepreneurship," *AEA Papers and Proceedings*, 111, 582–86.

——— (2024): "Common venture capital investors and startup growth," *The Review of Financial Studies*, 37, 549–590.

Emmens, J., D. Hutschenreiter, S. Manfredonia, F. Noth, and T. Santini (2024): "From shares to machines: How common ownership drives automation," Tech. rep., IWH Discussion Papers.

Ewens, M. and N. Malenko (2020): "Board Dynamics over the Startup Life Cycle," Working Paper 27769, National Bureau of Economic Research.

Faccio, M. and W. J. O'Brien (2021): "Business groups and employment," *Management Science*, 67, 3468–3491.

Gibson, M. (2024): "How Major Tech Firms Used Illegal "No-Poach" Agreements to Control Workers' Salaries," Tech. rep., WE Upjohn Institute for Employment Research.

Giroud, X. and H. M. Mueller (2015): "Capital and Labor Reallocation within Firms," *The Journal of Finance*, 70, 1767–1804.

Gompers, P. A., W. Gornall, S. N. Kaplan, and I. A. Strebulaev (2020): "How do venture capitalists make decisions?" *Journal of Financial Economics*, 135, 169–190.

González-Uribe, J. (2020): "Exchanges of innovation resources inside venture capital portfolios," *Journal of Financial Economics*, 135, 144–168.

Goodman-Bacon, A. (2021): "Difference-in-differences with variation in treatment timing," *Journal of Econometrics*, 225, 254–277.

Granovetter, M. S. (1973): "The strength of weak ties," *American Journal of Sociology*, 78, 1360–1380.

Gulati, R. (1998): "Alliances and networks," *Strategic Management Journal*, 19, 293–317.

Gutiérrez, G. and T. Philippon (2016): "Investment-less growth: An empirical investigation," Tech. rep., National Bureau of Economic Research.

Hacamo, I. and K. Kleiner (2022): "Forced entrepreneurs," *The Journal of Finance*, 77, 49–83.

Hellmann, T. and M. Puri (2002): "Venture capital and the professionalization of start-up firms: Empirical evidence," *The Journal of Finance*, 57, 169–197.

Hensvik, L. and O. N. Skans (2016): "Social networks, employee selection, and labor market outcomes," *Journal of Labor Economics*, 34, 825–867.

Herrera-Caicedo, A., J. Jeffers, and E. Prager (2025): "Collusion through Common Leadership," Working Paper 33866, National Bureau of Economic Research.

Huneeus, F., B. Larrain, M. Larrain, and M. Prem (2021): "The internal labor markets of business groups," *Journal of Corporate Finance*, 69, 102017.

Jeffers, J. S. (2023): "The Impact of Restricting Labor Mobility on Corporate Investment and Entrepreneurship," *The Review of Financial Studies*, 37, 1–44.

Johnson, M. S., K. Lavetti, and M. Lipsitz (2025): "The Labor Market Effects of Legal Restrictions on Worker Mobility," *Journal of Political Economy*, 133, 2735–2793.

Jovanovic, B. (1979): "Job matching and the theory of turnover," *Journal of political economy*, 87, 972–990.

Kortum, S. and J. Lerner (2000): "Assessing the contribution of venture capital," *the RAND Journal of Economics*, 31, 674–692.

Krueger, A. B. and O. Ashenfelter (2022): "Theory and evidence on employer collusion in the franchise sector," *Journal of Human Resources*, 57, S324–S348.

Kwan, A., B. Matthies, R. R. Townsend, and T. Xu (2025): "Entrepreneurial Spawning from Remote Work," Working Paper 33774, National Bureau of Economic Research.

Lafontaine, F., Saattvic, and M. E. Slade (2025): "No-poaching clauses in franchise contracts, anticompetitive or efficiency enhancing?" *Available at SSRN 4404155*.

Leccese, M. (2024): "Strategic Investment in Competitors: Theory and Evidence from Technology Startups," *Available at SSRN 4968068*.

Li, X., T. Liu, and L. A. Taylor (2023): "Common ownership and innovation efficiency," *Journal of Financial Economics*, 147, 475–497.

Lippman, S. A. and R. P. Rumelt (1982): "Uncertain imitability: An analysis of interfirm differences in efficiency under competition," *The Bell Journal of Economics*, 418–438.

Marx, M., D. Strumsky, and L. Fleming (2009): "Mobility, skills, and the Michigan non-compete experiment," *Management Science*, 55, 875–889.

Montag, F. (2023): "Mergers, foreign competition, and jobs: Evidence from the US appliance industry," *George J. Stigler Center for the Study of the Economy & the State Working Paper*.

Montgomery, J. D. (1991): "Social networks and labor-market outcomes: Toward an economic analysis," *The American economic review*, 81, 1408–1418.

——— (1992): "Job search and network composition: Implications of the strength-of-weak-ties hypothesis," *American Sociological Review*, 586–596.

Newham, M., J. Seldeslachts, and A. Banal-Estañol (2025): "Common Ownership and Market Entry: Evidence from the Pharmaceutical Industry," *American Economic Journal: Microeconomics*, 17, 260–327.

O'brien, D. P. and S. C. Salop (1999): "Competitive effects of partial ownership: Financial interest and corporate control," *Antitrust LJ*, 67, 559.

Pallais, A. and E. G. Sands (2016): "Why the referential treatment? Evidence from field experiments on referrals," *Journal of Political Economy*, 124, 1793–1828.

Prager, E. and M. Schmitt (2021): "Employer Consolidation and Wages: Evidence from Hospitals," *American Economic Review*, 111, 397–427.

Puri, M. and R. Zarutskie (2012): "On the Life Cycle Dynamics of Venture-Capital- and Non-Venture-Capital-Financed Firms," *The Journal of Finance*, 67, 2247–2293.

Rauterberg, G. and E. Talley (2017): "Contracting out of the fiduciary duty of loyalty: An empirical analysis of corporate opportunity waivers," *Colum. L. Rev.*, 117, 1075.

Rosenkopf, L. and P. Almeida (2003): "Overcoming local search through alliances and mobility," *Management Science*, 49, 751–766.

Rotemberg, J. (1984): "Financial transaction costs and industrial performance," Tech. rep., Sloan School of Management Working paper; 1554-84.

Samila, S. and O. Sorenson (2011): "Venture capital, entrepreneurship, and economic growth," *The Review of Economics and Statistics*, 93, 338–349.

Shapiro, C. (2019): "Protecting Competition in the American Economy: Merger Control, Tech Titans, Labor Markets," *Journal of Economic Perspectives*, 33, 69–93.

Shi, K. (2024): "Mergers and Mismatches in the Labor Market for Creativity," *Working Paper*.

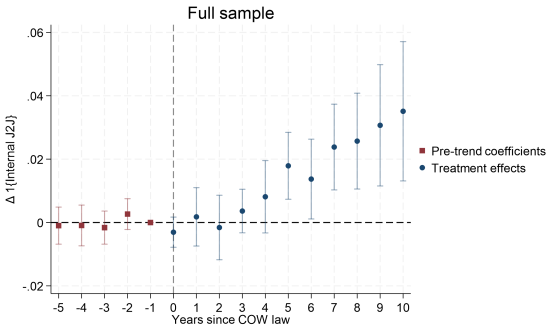
Stolpe, M. (2002): "Determinants of knowledge diffusion as evidenced in patent data: the case of liquid crystal display technology," *Research Policy*, 31, 1181–1198.

Tate, G. and L. Yang (2015): "The Bright Side of Corporate Diversification: Evidence from Internal Labor Markets," *The Review of Financial Studies*, 28, 2203–2249.

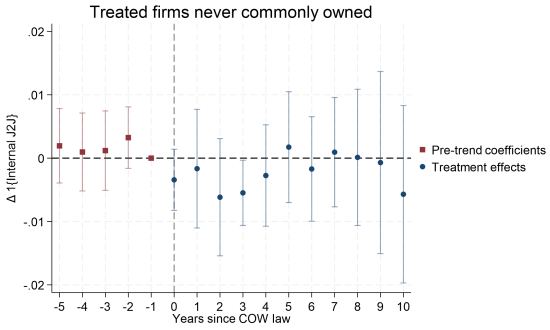
——— (2024): "The human factor in acquisitions: Cross-industry labor mobility and corporate diversification," *The Review of Financial Studies*, 37, 45–88.

ONLINE APPENDICES

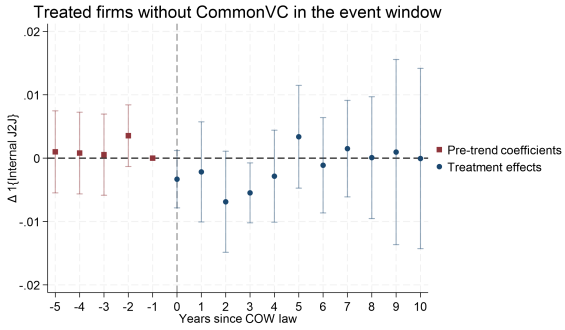
A. Exclusion placebo



A. Full sample



B. Treated firms never commonly owned



C. Treated firms without CommonVC in the event window

FIGURE A.1. Exclusion placebo: COW adoption and within-market mobility in the full sample and at ventures without common ownership

Notes: The figure tests the exclusion restriction behind the COW instrument: if the adoption of a corporate opportunity waiver (COW) statute affects within-market mobility only by creating common ownership, ventures that never become commonly owned should show no change in mobility after adoption. The sample follows the stacked COW design: for each adoption cohort g (Delaware 2000, Oklahoma 2001, Missouri 2003, Kansas 2005, Texas 2006, Nevada 2007, New Jersey 2011, Maryland 2014, Washington 2016), treated firms are ventures incorporated in the state that adopts a COW statute in year g and control firms are ventures incorporated in states that never adopt one, observed in years $g - 5$ to $g + 10$; the cohort data sets are stacked. Panel (a) uses all treated and control ventures. Panels (b) and (c) restrict the treated group to ventures without common ownership, while all control firms are kept: in panel (b), treated ventures with $CommonVC = 0$ in every year in which they appear in the panel; in panel (c), treated ventures with $CommonVC = 0$ in every year of their cohort's event window, $g - 5$ to $g + 10$. Vertical bars are 95% confidence intervals; standard errors clustered at the state-of-incorporation level.

TABLE A.1. Exclusion placebo: COW adoption and within-market mobility at ventures without common ownership

	(1) Treated firms never commonly owned	(2) Treated firms without CommonVC in the event window
Average post-adoption effect ($k = 0, \ldots, 10$)	-0.00225	-0.00145
	(0.00427)	(0.00394)
p-value	0.601	0.714
Pre-trend p-value ($k = -5, \ldots, -2$)	0.697	0.519
MDE (80% power)	0.0122	0.0113
Observations	2,209,407	2,229,631
Treated firm-years	100,346	120,570
Mean DV	0.0343	0.0343
Treated pre-adoption mean	0.0148	0.0151

Notes: This table reports a placebo test of the exclusion restriction behind the COW instrument: if the adoption of a corporate opportunity waiver (COW) statute affects within-market mobility only by creating common ownership, ventures that never become commonly owned should show no change in mobility after adoption. The sample follows the stacked COW design: for each adoption cohort g (Delaware 2000, Oklahoma 2001, Missouri 2003, Kansas 2005, Texas 2006, Nevada 2007, New Jersey 2011, Maryland 2014, Washington 2016), treated firms are ventures incorporated in the state that adopts a COW statute in year g and control firms are ventures incorporated in states that never adopt one, observed in years $g-5$ to $g+10$; the cohort data sets are stacked. The treated group is restricted to ventures without common ownership, while all control firms are kept: in column (1), treated ventures with $CommonVC = 0$ in every year in which they appear in the panel; in column (2), treated ventures with $CommonVC = 0$ in every year of their cohort's event window, $g-5$ to $g+10$. The dependent variable is regressed on indicators for treated firms in each event year $k = t - g$, $k = -5, \ldots, 10$, relative to the year before adoption ($k = -1$), with cohort-by-firm, cohort-by-year, industry-by-year and CBSA-by-year fixed effects and no controls. Standard errors clustered at the state-of-incorporation level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B. Details on Industry Crosswalk Construction

This appendix describes the construction of the industry crosswalk between PitchBook and VentureXpert. The purpose of the crosswalk is to harmonize industry classifications across the two data sources. Because PitchBook and VentureXpert use different industry taxonomies, we map each VentureXpert industry category to the closest corresponding PitchBook industry category.

We construct the crosswalk at the broad industry level. Table B.1 reports the resulting mapping. The first column lists PitchBook industries, and the second column lists the corresponding VentureXpert industries. The mapping is many-to-one because VentureXpert often uses more granular categories than PitchBook. Thus, multiple VentureXpert industries may map to the same PitchBook industry.

The matching procedure proceeds in three steps. First, we standardize industry labels across the two data sources. We harmonize common abbreviations and closely related terms, remove punctuation, and remove generic words that do not distinguish industries, such as "other" and "related." Second, we compare each VentureXpert industry category with candidate PitchBook industries based on semantic similarity between industry labels and the underlying economic activities they describe. Third, we assign each VentureXpert industry to the PitchBook industry that best captures its primary activity.

In most cases, the mapping is straightforward. For example, VentureXpert com munication-related categories such as *Data Comm.*, *Wireless Communications*, *Satellite Comm*, and *Telephone Rel.* are mapped to PitchBook's *Communications and Networking*. VentureXpert biotechnology-related categories such as *Biotech Research*, *Biotech-Human*, *Biotech-Animal*, and *Biotech-Industrial* are mapped to PitchBook's *Pharmaceuticals and Biotechnology*. Medical diagnostics, biosensors, and medical/health products are mapped to PitchBook's *Healthcare Devices and Supplies*, while medical/health services are mapped to PitchBook's *Healthcare Services*.

For ambiguous categories, we manually review the automated match and assign the VentureXpert category to the PitchBook industry that best reflects its substantive economic activity. For instance, internet-related categories are classified based on whether they primarily correspond to software, retail, communications infrastructure, or media content. This procedure preserves the VentureXpert industry information while allowing VentureXpert observations to be classified consistently using the PitchBook industry taxonomy.

TABLE B.1. Pitchbook and VenturExpert Industry Crosswalk

Pitchbook Industry	VenturExpert Industry
Agriculture	Agricultural, Forestry
Chemicals and Gases	Chemicals and Materials
Commercial Products	Scientific Instrumentation
Commercial Products	Industrial Products, Other
Commercial Products	Manufacturing
Commercial Products	Industrial Automation
Commercial Products	Laser Related
Commercial Products	Electronics Equipment
Commercial Products	Industrial Equipment
Commercial Services	Business Services
Commercial Services	Industrial Services
Communications and Networking	Telephone Rel.
Communications and Networking	Data Comm.
Communications and Networking	Satellite Comm
Communications and Networking	Fiber Optics
Communications and Networking	Wireless Communications
Communications and Networking	Internet Communications
Communications and Networking	Facsimile Trans
Communications and Networking	Internet Services
Communications and Networking	Comm. Other
Communications and Networking	Commer. Comm.
Computer Hardware	Computers Hardware
Computer Hardware	Computer Peripherals
Construction (Non-Wood)	Construction
Consumer Non-Durables	Food and Beverage
Energy Equipment	Power Supplies
Energy Equipment	Batteries
Energy Services	Energy, Conservation

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TABLE B.1 – *Continued from previous page*

Pitchbook Industry	VenturExpert Industry
Energy Services	Energy, Enhanced Recovery
Exploration, Production and Refining	Oil & Gas Exploration
Healthcare Devices and Supplies	Biotech Equipment
Healthcare Devices and Supplies	Medical Diagnostics
Healthcare Devices and Supplies	Biosensors
Healthcare Devices and Supplies	Med/Health Products
Healthcare Services	Med/Health Services
IT Services	Internet Programming
IT Services	Computer Services
IT Services	Computer Programming
IT Services	Turnkey Integrated Systems and Solutions
Media	Internet Content
Other Business Products and Services	Other
Other Consumer Products and Services	Consumer Services
Other Consumer Products and Services	Consumer, Other
Other Consumer Products and Services	Consumer Products
Other Energy	Energy, Alternative
Other Energy	Energy, Coal
Other Energy	Energy, Other
Other Financial Services	Financial Services
Other Information Technology	Computer Other
Other Information Technology	Electronics, Other
Other Materials	Pollution and Recycling
Pharmaceuticals and Biotechnology	Biotech-Industrial
Pharmaceuticals and Biotechnology	Biotech Research
Pharmaceuticals and Biotechnology	Biotech Other
Pharmaceuticals and Biotechnology	Pharmaceutical
Pharmaceuticals and Biotechnology	Biotech-Animal
Pharmaceuticals and Biotechnology	Biotech-Human

Continued on next page

TABLE B.1 – *Continued from previous page*

Pitchbook Industry	VenturExpert Industry
Pharmaceuticals and Biotechnology	Medical Therapeutics
Restaurants, Hotels and Leisure	Entertainment and Leisure
Retail	Retailing Related
Retail	Internet Ecommerce
Semiconductors	Semiconductors/Other Electronics
Semiconductors	Optoelectronics
Software	Internet Software
Software	E-Commerce Technology
Software	Computer Software
Software	Digital Imaging and Computer Graphics
Transportation	Transportation
Utilities	Utilities

Notes: This table reports the industry crosswalk between PitchBook and VentureXpert. The first column reports the PitchBook industry category, and the second column reports the corresponding VentureXpert industry category. The crosswalk is many-to-one because VentureXpert often uses more granular categories than PitchBook. Therefore, multiple VentureXpert industries may map to the same PitchBook industry.

C. Additional Figures and Tables

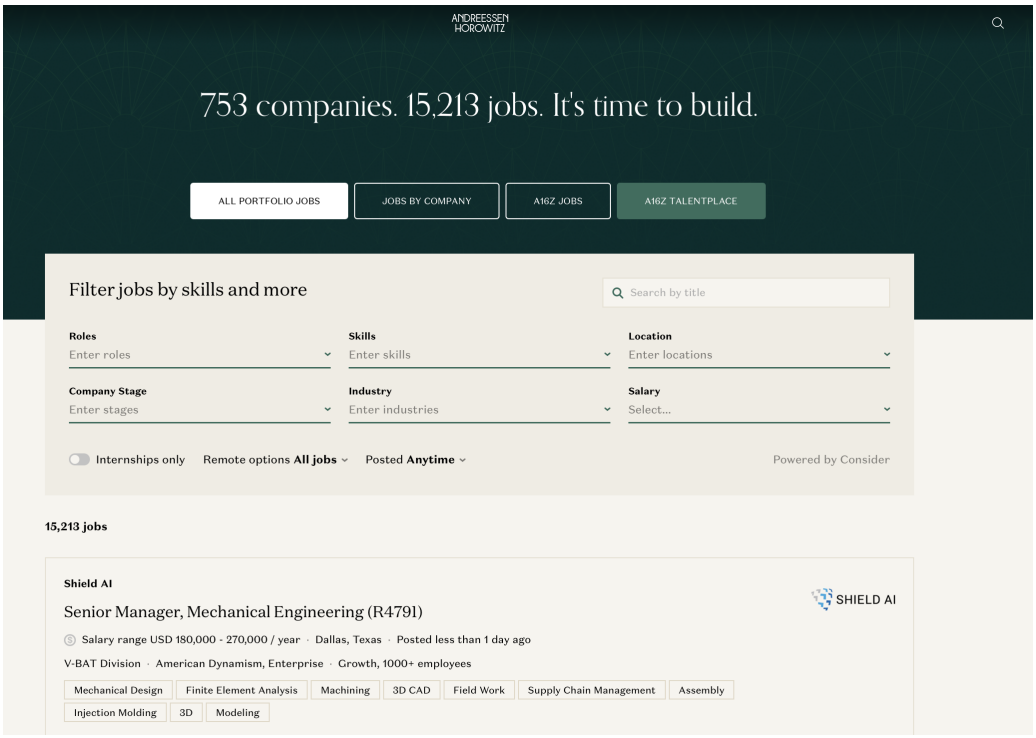
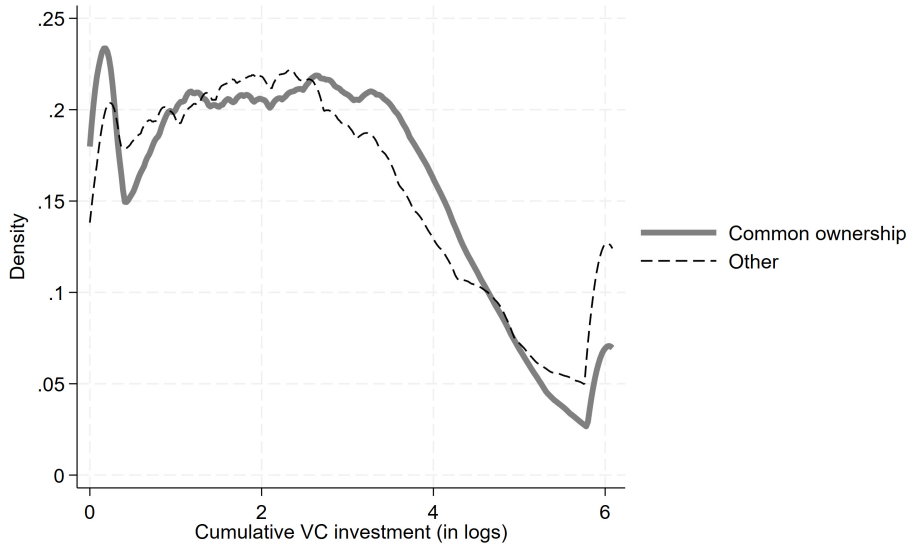


FIGURE C.1. A venture capital firm's portfolio jobs board

Notes: Screenshot of the Andreessen Horowitz (a16z) portfolio jobs board, accessed in April 2026, available at https://jobs.a16z.com/jobs. The board aggregates open positions across a16z portfolio companies and allows candidates to filter by industry, company stage, location, skills, and salary. Headline counts of listed companies and open positions reflect the date of access and change over time.

FIGURE C.2. Distribution of cumulative VC investment, by common ownership status



Notes: This figure plots kernel density estimates of cumulative venture capital investment (in logs) for ventures in the sample, separately by common ownership status. The solid grey line corresponds to venture-years where at least one common VC investor is observed (common ownership); the black dashed line corresponds to all other venture-years. Cumulative VC investment is measured in millions of USD and transformed as ln(1 + cumulative VC investment). The sample is restricted to ventures with strictly positive cumulative VC investment. Densities are estimated using an Epanechnikov kernel.

TABLE C.1. Summary of Corporate Opportunity Waiver Timeline

State incorp	Effective Date	By Charter	By bylaws	By Board action	Covers directors	Covers Shareholders
DE	7/1/2000	Yes	Possible	Yes	Yes	Yes
OK	11/1/2001	Yes	Possible	Yes	Yes	Yes
MO	10/1/2003	Yes	Possible	Yes	Yes	Yes
KS	1/1/2005	Yes	Possible	Yes	Yes	Yes
TX	1/1/2006	Yes	Possible	Yes	Yes	Yes
NV	10/1/2007	Yes	Possible	Yes	Yes	Yes
NJ	3/11/2011	Yes	Possible	Yes	Yes	Yes
MD	10/1/2014	Yes	Possible	Yes	Yes	No
WA	1/1/2016	Yes	No	No	Yes	Yes

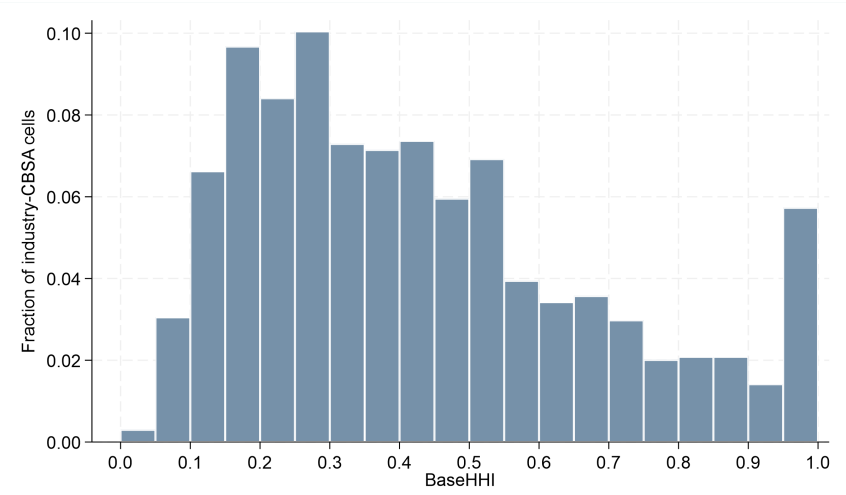
Note: This table provides the summary information of the states that amended their corporate laws explicitly to allow COWs (Source: Rauterberg and Talley (2017)).

TABLE C.2. Common ownership and the number of within-market moves by worker type: PPML

	Number of internal J2J moves					
	(1) All workers	(2) Innovators	(3) C-suite	(4) Junior	(5) Mid-seniority	(6) Senior
CommonVC	0.719***	0.812***	0.990***	0.640***	0.843***	1.101***
	(0.0435)	(0.0608)	(0.129)	(0.0454)	(0.0463)	(0.113)
Observations	224,395	57,198	11,975	157,500	91,837	18,595
Pseudo R-squared	0.501	0.307	0.175	0.409	0.340	0.179
Mean DV	0.478	0.255	0.133	0.298	0.247	0.135
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
CBSA-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Poisson pseudo-maximum-likelihood estimates. The dependent variable is the number of workers of the column's type who make an internal J2J move in the year (definitions in Table 1). Firm-years in fixed-effect groups with only zero outcomes are dropped, so the number of observations differs across columns. Investor controls as in Table 2. Standard errors clustered at the industry-CBSA level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

FIGURE C.3. Distribution of *BaseHHI*



Notes: This figure plots the distribution of pre-period local labor market concentration across industry-CBSA cells. Concentration is measured by the Herfindahl-Hirschman index, $\text{PreHHI}_{n,c} = \sum_j s^2_{j,\tau-1}$, where $s_{j,\tau}$ is firm j's employment share in industry-CBSA cell (n,c) in year τ, the first year the cell exhibits any common-VC ownership.

TABLE C.3. Robustness analysis on within-portfolio worker mobility

	$\mathbb{1}\{\text{Mobility}\} \times 10^4$					
	(1) All workers	(2) All workers	(3) Innovators	(4) Innovators	(5) C-suites	(6) C-suites
SharedVC	16.108***	10.716***	3.291***	2.295***	0.469***	0.311***
	(1.760)	(1.236)	(0.215)	(0.155)	(0.113)	(0.076)
SameInd × *SharedVC*		19.104***		3.530***		0.559***
		(4.277)		(0.807)		(0.164)
Observations	918,957,912	918,957,912	918,957,912	918,957,912	918,957,912	918,957,912
Mean DV	2.158	2.158	0.316	0.316	0.038	0.038
Origin-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Destination-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The unit of observation is a directed dyad-year (o, d, t): an ordered pair of distinct ventures, origin o and destination d, in the same CBSA in year t (1990–2021). $\mathbb{1}\{\text{Mobility}\}$ equals one if at least one worker's main job moves from o to d in year t (the main job at o ends in a month of t and the job at d becomes the worker's main job within 12 months; monthly main job built from the Revelio positions), and zero otherwise. Columns (3)–(4) count only movers with scientist or engineer titles and columns (5)–(6) only movers with C-suite titles (chief officers and general counsel), both at the end of the main job at o and at the start of the main job at d. The dependent variable is multiplied by 10^4, so coefficients, standard errors and Mean DV are multiplied by 10^4. *SharedVC* equals one if o and d share a VC investor in year t; *SameInd* equals one if they operate in the same industry. All regressions include origin-firm-by-year and destination-firm-by-year fixed effects instead of the dyad and year fixed effects of Table 4. They do not include *SameInd* on its own, which these fixed effects, unlike the dyad fixed effects, do not absorb; 4,724 singleton observations are dropped. Standard errors clustered by CBSA (367 clusters) in parentheses; p-values from t with 366 degrees of freedom. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.4. Within-portfolio worker mobility by job role

	(1) *SharedVC*	(2) *SharedVC*	(3) *SameInd* × *SharedVC*	Mean DV ($\times 10^4$)	Observations
	(coefficients and standard errors $\times 10^4$)				
By occupation					
Scientists	0.281***	0.102***	0.656*	0.0256	903,782,148
	(0.090)	(0.039)	(0.386)		
Multiplier	11.0×	4.0×	25.7×		
Engineers	3.114***	2.395***	2.633***	0.2900	903,782,148
	(0.201)	(0.195)	(0.456)		
Multiplier	10.7×	8.3×	9.1×		
Sales	3.688***	2.057***	5.980***	0.3235	903,782,148
	(0.320)	(0.199)	(0.842)		
Multiplier	11.4×	6.4×	18.5×		
Other workers	9.181***	6.470***	9.937***	1.1762	903,782,148
	(1.066)	(0.719)	(3.082)		
Multiplier	7.8×	5.5×	8.4×		
CEOs	0.058**	0.035*	0.086	0.0064	903,782,148
	(0.029)	(0.019)	(0.097)		
Multiplier	9.1×	5.4×	13.5×		
By seniority					
Junior	8.446***	5.972***	9.068***	1.1147	903,782,148
	(0.837)	(0.581)	(2.316)		
Multiplier	7.6×	5.4×	8.1×		
Mid-seniority	4.794***	2.926***	6.846***	0.4577	903,782,148
	(0.542)	(0.266)	(1.548)		
Multiplier	10.5×	6.4×	15.0×		
Senior	0.659***	0.410***	0.913**	0.0535	903,782,148
	(0.182)	(0.103)	(0.388)		
Multiplier	12.3×	7.7×	17.1×		
Dyad FE	Yes	Yes	Yes		
Year FE	Yes	Yes	Yes		

Notes: Each observation is a directed pair of ventures (origin o, destination d) in the same CBSA in year t (1990–2021). The dependent variable is $\mathbb{1}\{\text{Mobility}\}$ for the job role of the row: one if at least one worker in that role moves from o to d in t, zero otherwise. Moves are main-job job-to-job moves: the worker's main job (one per month, from the Revelio positions) at o ends in a month of t and the job at d becomes the main job within 12 months. A move has a role when the worker's most senior position at o in the exit month and at d in the month it becomes the main job both have that role. Scientists and engineers: scientist and engineer titles; sales: sales, marketing, business-development, account-management, partnerships, growth, revenue and customer-success titles that are not scientist or engineer titles; other workers: titles that are none of these, including positions without a title; CEOs: chief executive officer titles; junior, mid-seniority and senior: Revelio seniority levels 1–3, 4–5 and 6–7. Roles are not mutually exclusive. Innovators and C-suite executives are in Table 4. Column (1): regression on the first regressor alone; columns (2) and (3): one regression on both, as columns (1) and (2) of Table 4. The dependent variable is multiplied by 10^4: coefficients, standard errors and Mean DV are multiplied by 10^4. *Multiplier*: coefficient divided by the mean of the dependent variable in the estimation sample (singletons dropped), to which Mean DV and Observations refer. *SharedVC* equals one if o and d share a VC investor in t; *SameInd* equals one if they operate in the same industry. All regressions include dyad and year fixed effects, as in Table 4. Standard errors clustered by CBSA (445 clusters) in parentheses; p-values from t with 444 degrees of freedom. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.5. Separations and worker-specific choice sets

	Number or share
Panel A: Separations (worker–origin firm–year of separation)	
Separations from ventures in the sample, 1990–2021	10,786,148
Separations from a venture active in the sample in the year of separation (baseline)	10,087,415
share with a subsequent job	0.858
share reemployed within 36 months	0.656
share reemployed within 36 months, observed windows only	0.800
share reemployed at a venture in the choice set	0.030
share reemployed at a venture in the choice set within 36 months	0.023
share with the 36-month window observed	0.820
share with *CommonVC* at the origin	0.274
Panel B: Choice sets (origin–potential destination–year of separation), baseline sample	
Origin firm–year of separation cells	488,967
Origin–potential destination–year observations	584,319,582
Worker–potential destination observations represented	12,897,620,455
Choice-set size: mean / median across origin–year cells	1,195 / 796
Choice-set size: mean / median across separations	1,279 / 879
Choice-set size: maximum	4,773
Share of origin–destination pairs with *SharedVC*	0.0134
Share of origin–destination pairs with *SameInd*	0.141
Realized moves to a venture in the choice set within 36 months	229,706

Notes: A separation is a Revelio employment spell (v6 worker file, spells built from position dates) ending at a venture in our sample; the year of separation is the year of the spell's last end date. In the v6 worker file the spell's last year in the position panel equals the year of its last end date for every spell, so no separation is flagged as an ongoing position. The baseline sample, used for the shares in Panel A, for Panel B and for the regressions, keeps the 10,087,415 separations from ventures active in the sample (private VC-backed ventures of the firm-year panel) in the year of separation; the shares in Panel A are of these separations, not of all 10,786,148. A next job is observed within 36 months only when the 36-month window ends by December 2021 (separations up to 2018); the observed-window share conditions on those separations. The choice set of a worker separating from origin i in year t consists of every other venture active in our sample in year t and located in the origin's CBSA, in any industry; 1,180 origin–year cells (0.24 percent) have an empty choice set and enter the cell-level mean and median with size zero. Worker–potential destination observations are separations times choice-set size, summed over origin–year cells. A pair of ventures has *SharedVC* equal to one when at least one VC investor holds both in year t (holding period from the investor's first deal year in the venture to five years after the venture's last deal, or to its exit year if that is earlier) and *SameInd* equal to one when they operate in the same industry; both shares are over origin–potential destination–year observations. Realized moves are separations of the baseline sample reemployed at a venture of the choice set within 36 months.

TABLE C.6. Shared investors and the destination of separating workers, by horizon

Next job within	(1) 6 months	(2) 12 months	(3) 18 months	(4) 36 months	(5) any time
Panel A: Workers reemployed at a venture in the choice set within the horizon					
SharedVC	10.686***	10.580***	10.702***	11.834***	10.354***
	(3.880)	(3.828)	(3.829)	(4.149)	(3.679)
SameInd	10.748***	10.837***	10.962***	11.896***	10.914***
	(2.804)	(2.798)	(2.800)	(2.888)	(2.797)
Observations	499,321,531	532,782,059	528,917,164	442,023,962	633,975,162
Mean DV ($\times$10,000)	4.722	4.781	4.840	5.195	4.803
Multiplier	2.3$\times$	2.2$\times$	2.2$\times$	2.3$\times$	2.2$\times$
Panel B: All separations with the horizon window observed					
SharedVC	1.136***	1.248***	1.311***	1.481***	1.223***
	(0.300)	(0.324)	(0.333)	(0.346)	(0.319)
SameInd	0.595***	0.661***	0.688***	0.737***	0.719***
	(0.097)	(0.106)	(0.109)	(0.110)	(0.117)
Observations	12,396,596,633	11,980,862,868	11,544,483,001	9,872,740,625	12,897,619,797
Mean DV ($\times$10,000)	0.190	0.213	0.222	0.233	0.236
Multiplier	6.0$\times$	5.9$\times$	5.9$\times$	6.4$\times$	5.2$\times$

Notes: The unit of observation is a worker–origin firm–potential destination firm–year-of-separation combination (w, i, j, t), where the potential destinations are a worker-specific choice set consisting of all VC-backed ventures active in the origin firm's CBSA in the year of separation, in any industry. The dependent variable equals one if the worker moves from the origin to that destination within 36 months and zero otherwise, including all zeros for workers with no subsequent job and for workers whose next job is outside the choice set; coefficients, standard errors and means are multiplied by 10,000. *SharedVC* equals one if the origin and the potential destination share a VC investor in the year of separation, and *SameInd* equals one if they operate in the same industry. Estimation uses the origin–destination–year cell means weighted by the number of separations, which is numerically identical to a worker-level regression. Panel A conditions on workers reemployed at a venture in the choice set within the horizon; Panel B uses all separations from active ventures whose horizon window is observed (for any time, all separations). All columns include origin-year and destination fixed effects. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.7. Shared investors and the destination of separating workers, by worker group

Workers	(1) Innovators	(2) High seniority	(3) C-suites
Panel A: Workers reemployed at a venture in the choice set within 36 months			
SharedVC	8.660***	18.424***	21.194***
	(2.572)	(6.433)	(7.521)
SameInd	9.024***	11.596***	11.355***
	(2.513)	(2.696)	(2.592)
Observations	81,074,113	23,760,249	12,856,786
Mean DV ($\times$10,000)	4.986	5.053	5.154
Multiplier	1.7$\times$	3.6$\times$	4.1$\times$
Panel B: All separations of the group, 36-month window observed			
SharedVC	1.562***	2.857***	3.461***
	(0.313)	(0.725)	(0.892)
SameInd	0.939***	1.043***	1.249***
	(0.179)	(0.191)	(0.245)
Observations	1,021,786,947	337,114,588	139,917,811
Mean DV ($\times$10,000)	0.396	0.357	0.477
Multiplier	3.9$\times$	8.0$\times$	7.3$\times$

Notes: The unit of observation is a worker–origin firm–potential destination firm–year-of-separation combination (w, i, j, t), where the potential destinations are a worker-specific choice set consisting of all VC-backed ventures active in the origin firm's CBSA in the year of separation, in any industry. The dependent variable equals one if the worker moves from the origin to that destination within 36 months and zero otherwise, including all zeros for workers with no subsequent job and for workers whose next job is outside the choice set; coefficients, standard errors and means are multiplied by 10,000. *SharedVC* equals one if the origin and the potential destination share a VC investor in the year of separation, and *SameInd* equals one if they operate in the same industry. Estimation uses the origin–destination–year cell means weighted by the number of separations, which is numerically identical to a worker-level regression. Worker groups are defined by the position held at the origin: innovators are scientists and engineers, high-seniority workers are those in the top seniority levels, and C-suites are senior executives. Panel A conditions on workers of the group reemployed at a venture in the choice set within 36 months; Panel B uses all separations of the group with the 36-month window observed. All columns include origin-year and destination fixed effects. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.8. Heterogeneity by venture size

Size measure	(1) Capital raised	(2) Capital raised	(3) Employment	(4) Employment
VARIABLES	𝟙{next job in 36m}	𝟙{next job in choice set}	𝟙{next job in 36m}	𝟙{next job in choice set}
Panel A: Extensive margin, by origin size				
CommonVC × *Small*	-0.0019	0.0041***	-0.0036	0.0168***
	(0.0023)	(0.0010)	(0.0035)	(0.0018)
CommonVC × *Large*	-0.0004	0.0046***	-0.0007	0.0044***
	(0.0021)	(0.0011)	(0.0017)	(0.0007)
Large	0.0012	0.0009*	0.0231***	0.0058***
	(0.0018)	(0.0005)	(0.0019)	(0.0015)
Observations	8,268,890	8,268,890	8,268,890	8,268,890
Mean DV	0.800	0.028	0.800	0.028
p-value: *Small* = *Large*	0.632	0.776	0.414	0.000
Origin FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Panel B: Destination choice among ventures in the choice set				
Interaction with Size measure	(1) Origin size Capital raised	(2) Origin size Employment	(3) Destination size Capital raised	(4) Destination size Employment
SharedVC × *SmallOrigin*	48.297**	25.816***		
	(23.743)	(9.872)		
SharedVC × *LargeOrigin*	10.434***	11.422***		
	(3.270)	(3.971)		
SharedVC			25.224*	2.188***
			(13.757)	(0.728)
SharedVC × *LargeDestination*			-14.548	14.383***
			(10.501)	(4.976)
LargeDestination			1.715***	3.092***
			(0.343)	(0.640)
SameInd	11.891***	11.895***	11.894***	11.883***
	(2.885)	(2.888)	(2.888)	(2.885)
Observations	442,023,962	442,023,962	442,023,962	442,023,962
Mean DV (×10,000)	5.195	5.195	5.195	5.195
p-value: *Small* = *Large*	0.068	0.017		
Origin-Year FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes

Notes: Venture size is measured in the year of separation. *Capital raised* is the cumulative disclosed size of the venture's VC rounds up to that year, and *Employment* is its Revelio headcount; *Large* denotes ventures above the within-year median across all active ventures in our sample, and *Small* those at or below it (for capital raised, ventures with any disclosed funding when the median is zero). Panel A is estimated at the worker–origin firm–year-of-separation level on separations from active ventures with the 36-month window observed; because size is measured in the year of separation and changes over time within an origin, *Large* enters on its own alongside the two *CommonVC* interactions. In Panel B, origin size is absorbed by the origin-year fixed effects in columns (1)–(2), and columns (3)–(4) include *LargeDestination* on its own. Panel B is the destination-choice sample of Table 5, Panel B (coefficients multiplied by 10,000): columns (1)–(2) interact *SharedVC* with origin size, and columns (3)–(4) with the size of the potential destination. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.9. Redeployment from contracting ventures, by worker group

Workers	(1) All	(2) Innovators	(3) High seniority	(4) C-suites
Panel A: Destination choice, ventures with an employment decline of at least 50%				
SharedVC	13.407***	10.022***	19.236***	21.550***
	(3.350)	(1.955)	(4.953)	(6.437)
SameInd	10.122***	10.025***	8.741***	8.343***
	(2.112)	(2.299)	(1.831)	(1.814)
Observations	22,505,643	5,870,627	1,800,066	1,448,447
Mean DV (×10,000)	4.919	4.911	4.739	4.708
Multiplier	2.7×	2.0×	4.1×	4.6×

Panel B: Transition speed, ventures with an employment decline of at least 50%

Workers	(1) All	(2) All	(3) Innovators	(4) Innovators
VARIABLES	𝟙{next job in 6m}	log days	𝟙{next job in 6m}	log days
SharedVC	0.032	-0.079	-0.017	-0.943
	(0.025)	(0.131)	(0.101)	(0.732)
SameInd	0.015	0.011	-0.232*	2.226***
	(0.036)	(0.285)	(0.121)	(0.602)
Observations	6,583	6,583	1,204	1,204
Mean DV	0.829	2.700	0.874	2.501

Panel C: Transition speed, all origins, by worker group

Workers	(1) Innovators	(2) Innovators	(3) High seniority	(4) High seniority
VARIABLES	𝟙{next job in 6m}	log days	𝟙{next job in 6m}	log days
SharedVC	0.020**	-0.037	-0.141*	1.567
	(0.009)	(0.089)	(0.080)	(1.117)
SameInd	0.012	0.076	0.156	-2.227**
	(0.009)	(0.116)	(0.139)	(0.838)
Observations	21,678	21,678	2,431	2,431
Mean DV	0.851	2.236	0.844	2.197

Notes: Panel A repeats the destination-choice regressions of Table 5, Panel B, for workers separating from ventures experiencing an employment decline of at least 50 percent, for all workers and by the position held at the origin; coefficients are multiplied by 10,000 and all columns include origin-year and destination fixed effects. Panels B and C repeat the transition-speed regressions: Panel B for the same contracting ventures, with origin-year and destination fixed effects for all workers and origin, year and destination fixed effects for innovators; Panel C for all origins by worker group. High-seniority and C-suite movers from contracting ventures are too few (139 and 133) for *SharedVC* to vary within the fixed-effect cells, as are C-suite movers overall (1,161), so those speed regressions are omitted. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.10. Redeployment from contracting ventures, by worker group

	(1) All Workers	(2) Innovators	(3) C-suites	(4) Junior	(5) Mid-seniority	(6) High seniority
SharedVC	13.407***	10.022***	21.550***	11.355***	14.812***	19.236***
	(3.350)	(1.955)	(6.437)	(2.797)	(3.718)	(4.953)
SameInd	10.122***	10.025***	8.343***	10.176***	10.168***	8.741***
	(2.112)	(2.299)	(1.814)	(2.157)	(2.199)	(1.831)
Observations	22,505,643	5,870,627	1,448,447	13,456,394	8,426,723	1,800,066
Movers	11,071	2,883	682	6,487	4,192	853
Mean DV (×10,000)	4.919	4.911	4.708	4.821	4.975	4.739
Multiplier	2.7×	2.0×	4.6×	2.4×	3.0×	4.1×

Notes: The sample consists of workers separating from ventures whose employment falls by at least 50% between the year of separation and the venture's next year in the firm-year panel and who are reemployed within 36 months at a venture in the origin's CBSA choice set (all VC-backed ventures active in the CBSA in the year of separation), with the 36-month window observed. Separations and moves are spell-based (v6): the worker's next Revelio spell starts at the destination within 36 months of the end of the origin spell. Each column restricts the workers to the group, by any position held in the origin spell (raw titles; groups can overlap): innovators (scientists and engineers), C-suite executives, and Revelio seniority levels 1–3 (junior), 4–5 (mid-seniority) and 6–7 (high seniority). OLS of the worker–potential destination indicator for a move within 36 months, estimated on the origin–destination–year cell means weighted by the number of the group's movers from the origin-year (numerically identical to the worker-level regression); observations are worker–potential destination pairs; coefficients, standard errors and means are multiplied by 10,000; *Multiplier* is the coefficient divided by the mean. All specifications include origin-by-year and destination fixed effects. Standard errors clustered at the CBSA level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.11. Redeployment from ventures after their IPO, by worker group

	(1) All Workers	(2) Innovators	(3) C-suites	(4) Junior	(5) Mid-seniority	(6) High seniority
SharedVC	12.594***	9.208***	18.938***	11.093***	13.956***	22.245***
	(3.736)	(2.694)	(6.463)	(3.374)	(4.443)	(6.996)
SameInd	15.767***	14.951***	15.523***	15.422***	15.432***	15.921***
	(3.963)	(3.098)	(3.432)	(4.273)	(3.644)	(3.786)
Observations	43,569,167	8,726,552	1,228,343	27,505,754	16,847,806	2,364,220
Movers	25,316	5,161	677	15,721	9,839	1,347
Mean DV (×10,000)	5.811	5.914	5.511	5.716	5.840	5.697
Multiplier	2.2×	1.6×	3.4×	1.9×	2.4×	3.9×

Notes: The sample consists of workers separating from ventures in or after the year of their IPO (the earlier of the PitchBook and VentureXpert IPO dates; the venture must be active in the firm-year panel in the year of separation) who are reemployed within 36 months at a venture in the origin's CBSA choice set (all VC-backed ventures active in the CBSA in the year of separation), with the 36-month window observed. Separations and moves are spell-based: the worker's next Revelio spell starts at the destination within 36 months of the end of the origin spell. Each column restricts the workers to the group, by any position held in the origin spell (raw titles; groups can overlap): innovators (scientists and engineers), C-suite executives, and Revelio seniority levels 1–3 (junior), 4–5 (mid-seniority) and 6–7 (high seniority). Observations are worker–potential destination pairs; coefficients, standard errors and means are multiplied by 10,000; *Multiplier* is the coefficient divided by the mean. The regressions include origin-by-year and destination fixed effects. Standard errors clustered at the CBSA level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.12. Follow-on financing by the shared investor and worker redeployment

	(1)	(2)	(3)	(4)
	Decline $\geq$ 50%		Decline $\geq$ 70%	
Sample	In choice set	All	In choice set	All
SharedVC $\times$ *FollowOn*	11.974***	2.046***	13.834***	2.569***
	(1.602)	(0.360)	(2.028)	(0.428)
SharedVC $\times$ *NoFollowOn*	13.935***	2.168***	13.831***	2.212***
	(4.474)	(0.512)	(4.622)	(0.545)
SameInd	10.123***	1.096***	11.158***	1.298***
	(2.113)	(0.217)	(2.335)	(0.247)
Observations	22,505,643	243,462,790	11,697,495	114,798,411
Mean DV ($\times$10,000)	4.919	0.456	5.022	0.514
p-value: *FollowOn* = *NoFollowOn*	0.645	0.755	0.999	0.392
Origin-Year FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes

Notes: The sample is restricted to workers separating from ventures experiencing an employment decline of at least 50 or 70 percent between the year of separation and the following year. A potential destination sharing a VC investor with the origin is classified as *FollowOn* when a shared investor invested again in the origin after the first year in which it held both ventures and up to one year after the separation, and as *NoFollowOn* otherwise. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.13. Common ownership at Series A, follow-on financing, and directional worker flows

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$\mathbb{1}\{$worker moves from i to $j\}$ ($\times$10,000)						
SharedA	22.695***						
	(2.365)						
SameInd	19.748***	19.732***		19.736***	19.742***	19.733***	19.586***
	(2.884)	(2.886)		(2.883)	(2.885)	(2.887)	(2.871)
SharedA × *FollowOn*(i, j)		51.544***	45.019***			51.460***	48.171***
		(6.516)	(5.977)			(6.525)	(6.192)
SharedA × *FollowOn*(i) only		24.626***	8.518**				23.277***
		(7.206)	(4.079)				(6.958)
SharedA × *FollowOn*(j) only		9.331***	16.270***				9.208***
		(1.101)	(2.225)				(1.165)
SharedA × *NoFollowOn*		12.120***					13.742***
		(1.031)					(1.007)
SharedA × *FollowOn*(i)				38.507***			
				(4.267)			
SharedA × *NoFollowOn*(i)				11.167***			
				(0.952)			
SharedA × *FollowOn*(j)					31.066***		
					(2.701)		
SharedA × *NoFollowOn*(j)					16.591***		
					(2.965)		
SharedA × *NotBothFunded*						14.674***	
						(2.296)	
Observations	38,162,506	38,162,506	35,250,348	38,162,506	38,162,506	38,162,506	38,161,852
Mean DV ($\times$10,000)	12.070	12.070	12.671	12.070	12.070	12.070	12.070
Multiplier		4.3×	3.6×			4.3×	4.0×
p-value: *FollowOn*(i) = *FollowOn*(j)		0.026					0.026
p-value: both = neither		0.000					0.000
p-value: *FollowOn* = *NoFollowOn*				0.000	0.000		
Origin FE	Yes	Yes	No	Yes	Yes	Yes	No
Destination FE	Yes	Yes	No	Yes	Yes	Yes	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Origin-Year FE	No	No	No	No	No	No	Yes
Destination-Year FE	No	No	No	No	No	No	Yes
Dyad FE	No	No	Yes	No	No	No	No

Notes: The unit of observation is a directed pair of ventures ($i \to j$) in a given year. Both ventures raised a Series A round, are active in our sample in the same CBSA-year, and enter the panel in the year in which both have completed their Series A round. The dependent variable equals one if at least one worker moves from i to j in year t; coefficients, standard errors and means are multiplied by 10,000. *SharedA* equals one if the two ventures have a common VC investor in their Series A rounds. *FollowOn*(i) and *FollowOn*(j) equal one if such an investor invested again in i or in j after its Series A round and by year t. Column (3) includes dyad fixed effects, so the follow-on categories are identified from the timing of follow-on rounds within a pair, relative to *SharedA* pairs without follow-on financing. *Multiplier* reports the coefficient divided by the dependent-variable mean within the estimation sample, expressing the estimate as a multiple of the baseline rate. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.14. Series A common ownership and directional flows of innovators and C-suite executives

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	Innovator move			C-suite move		
SharedA × *FollowOn*(i, j)	9.850***			1.078***		
	(2.011)			(0.269)		
SharedA × *FollowOn*(i) only	4.789***			0.849*		
	(1.141)			(0.466)		
SharedA × *FollowOn*(j) only	2.904***			0.515**		
	(0.562)			(0.212)		
SharedA × *NoFollowOn*	2.797***			0.372**		
	(0.469)			(0.174)		
SameInd	3.590***	3.591***	3.592***	0.468***	0.468***	0.468***
	(0.647)	(0.647)	(0.647)	(0.073)	(0.073)	(0.073)
SharedA × *FollowOn*(i)		7.397***			0.967***	
		(0.753)			(0.333)	
SharedA × *NoFollowOn*(i)		2.841***			0.423**	
		(0.480)			(0.164)	
SharedA × *FollowOn*(j)			6.481***			0.804***
			(0.882)			(0.216)
SharedA × *NoFollowOn*(j)			3.510***			0.541**
			(0.569)			(0.272)
Observations	38,162,506	38,162,506	38,162,506	38,162,506	38,162,506	38,162,506
Mean DV (×10,000)	2.530	2.530	2.530	0.229	0.229	0.229
p-value: *FollowOn*(i) = *FollowOn*(j)	0.083			0.358		
p-value: both = neither	0.001			0.006		
p-value: *FollowOn* = *NoFollowOn*		0.000	0.011		0.016	0.174
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: See Table C.13. The dependent variable equals one if at least one innovator, that is, a scientist or engineer (columns 1–3), or one C-suite executive (columns 4–6) moves from i to j in year t; coefficients, standard errors and means are multiplied by 10,000. All columns include origin, destination and year fixed effects. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE C.15. Post-transition outcomes at connected destinations: exit hazards with follow-up ending in 2020

	(1)	(2)	(3)	(4)
	Follow-up through 2021		Follow-up through 2020	
VARIABLES	Exit to non-employment	Exit to another job	Exit to non-employment	Exit to another job
SharedVC	-0.027	-0.023*	-0.028	-0.024**
	(0.027)	(0.012)	(0.027)	(0.012)
SameInd	-0.005	-0.041***	-0.004	-0.042***
	(0.019)	(0.010)	(0.019)	(0.010)
Spells	221,773	221,773	221,612	221,612
Events	26,986	119,740	26,986	115,945
Censored spells	194,787	102,033	194,626	105,667
Implied hazard ratio (*SharedVC*)	0.973	0.978	0.972	0.976
Entry seniority & occupation controls	Yes	Yes	Yes	Yes
Strata: Dest. industry × entry year	Yes	Yes	Yes	Yes

Notes: This table compares the cause-specific exit hazards of Table 8, columns (2) and (3), with a version in which follow-up ends on December 31, 2020. Columns (1) and (2) reproduce Table 8. In columns (3) and (4), departures up to December 31, 2020 are classified using the 2021 histories, so every classified departure has a full twelve-month window to observe whether a new position starts. Spells still in progress on December 31, 2020, including those that end in 2021, are censored on that date and keep their employment time through the cutoff; spells that begin after the cutoff are excluded. Censoring at the destination's acquisition or IPO is as in Table 8. All columns report Cox proportional-hazard estimates (Efron method for ties) with the baseline hazard stratified by destination-industry-by-entry-year cohort and control for entry seniority and occupation. Standard errors, clustered at the CBSA level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

References

Rauterberg, G. and E. Talley (2017): "Contracting out of the fiduciary duty of loyalty: An empirical analysis of corporate opportunity waivers," *Colum. L. Rev.*, 117, 1075.